

BanglaCyberBench: A Dual-Script Benchmark and a Script-Aware Ensemble for Bengali Cyberbullying Detection

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Abstract. Most Bengali cyberbullying detection has been done on a single dataset, in a single script, and evaluated only on a random split of that one dataset, which leaves the question a real moderation system faces unanswered: does a classifier trained on one collection still work on comments from somewhere else? We address this with two contributions. The first is BanglaCyberBench, a deduplicated benchmark of 94,323 unique comments that merges four public sources under one five-class taxonomy; it covers both Bangla script and Romanized Bangla (*Banglish*) in a single resource, a combination prior single-source corpora have not offered at this scale, with the label consolidation checked by a human agreement study. One scope note belongs up front. The labels type the harmful *content* of an isolated comment—generic abuse, sexual or gender-based harassment, religion-directed abuse, and threats—rather than cyberbullying in its interactional sense of repetition, power imbalance, or victim history, none of which the comment-level sources record. The second contribution is a deliberately lean, reproducible reference system: a script-aware ensemble of four transformer encoders in which a Bangla-only specialist is silenced on Romanized text, trained with a light adversarial recipe rather than a heavy stack of imbalance tricks. In-domain the ensemble reaches a Macro-F1 of about 0.82, clearing strong classical and zero-shot large-language-model references; and on the published single-source base paper’s own protocol it is competitive overall while clearly stronger on the rare, safety-critical *threat* class, which it lifts by roughly seven F1 points (0.7579 to 0.8292). The study’s more important finding is negative, and we think it is the most useful thing here: when a whole source is held out, Macro-F1 collapses to roughly 0.46–0.59, so in-domain accuracy badly overstates readiness on unseen data, and cross-distribution transfer—across sources, and by extension across scripts—is the open problem rather than in-domain accuracy. The benchmark, split-construction code, and evaluation scripts are released for reproduction, subject to the source licences.

Keywords: Bengali cyberbullying detection; Romanized Bangla; dual-script benchmark; script-aware ensemble; adversarial fine-tuning; class imbalance; cross-source robustness.

1. Introduction

Cyberbullying on social platforms is a daily reality for a large share of Bengali-speaking users, and the sheer volume of it puts manual moderation out of reach at any realistic staffing level. Bengali is spoken by more than 240 million people, yet it sits on the low-resource side of NLP: annotated corpora are small, they come from scattered sources with incompatible label schemes, and a large fraction of real abusive text is not even written in Bangla script. People routinely type Bengali in the Latin alphabet, so *tui akta faltu manush* and its Bangla-script equivalent carry the same insult in two different surface forms. A moderation model that only ever saw one of those forms is blind to the other, and most published Bengali cyberbullying work has looked at exactly one of them.

Three limitations run through the prior literature. Datasets are almost always single-source, so a model learns one platform’s quirks and one annotation team’s habits rather than the phenomenon itself. The script is almost always Bangla-only, so Romanized Bangla, which is ubiquitous in comments, is left untested. And evaluation almost always stops at a random split of the one corpus, so nothing tells us whether the model transfers. The strongest recent system, the transformer-stacking framework of Hoque et al. [1], reports 89.23% multiclass F1 on a 44,001-comment Facebook dataset, which is a genuinely strong single-source number, but it is a single-source number: one platform, one script, and no measurement of what happens off that distribution.

This paper is built around the gap those three limitations leave. We ask whether heterogeneous Bengali

cyberbullying data can be unified into one clean, deduplicated, dual-script, fine-grained benchmark; whether a script-aware model with a plain training recipe, reproducible end to end on one GPU, can classify abuse across both scripts without the heavy regularisation stack that is usually thrown at imbalanced text; and how far such a model actually travels when a whole source is held out. We answer all three, and we are deliberate about separating what is solved (in-domain, fine-grained classification), what is measured and found hard (cross-source transfer), and what the present source pool cannot yet isolate (cross-script transfer, which needs a second Romanized source before its number means anything).

Concretely, four research questions organise the study. **RQ1** (Section 3): can four incompatible Bengali sources be consolidated into one deduplicated, dual-script, five-class benchmark under a documented, human-checked taxonomy? **RQ2** (Sections 6.1 and 6.3): what in-domain performance does a lean, reproducible script-aware reference system reach on that benchmark, and where does it fail? **RQ3** (Sections 6.4, 6.5 and 6.8): which parts of the recipe (FGM, the script mask, the fusion rule) actually move the number, and by how much against seed noise? **RQ4** (Section 6.6): how far does in-domain accuracy overstate readiness when a whole source is held out? Each results subsection is tagged to the question it answers.

1.1 Contributions

(C1) A multi-source, dual-script, deduplicated benchmark. BanglaCyberBench merges four public Bengali datasets into 94,323 unique comments after removing 41,252 duplicates, covering Bangla script (60.4%) and Romanized Bangla (39.6%) under one schema. A hard uid assertion and a cleaned-text uniqueness audit are applied before every split.

(C2) A documented taxonomy with a human consistency check. Eighty-nine raw label strings from four incompatible schemas fold into five classes by an explicit priority rule that keeps explicit violence and identity-targeted abuse out of a generic bucket. Two of the authors independently re-labelled a stratified sample and agreed at Cohen’s $\kappa = 0.709$; we treat this as an internal consistency check on the scheme, not as external gold validation of all benchmark labels (Section 7.4).

(C3) A script-aware ensemble with a minimal recipe. Four encoders are fine-tuned with cross-entropy plus FGM only, and a Bangla-only specialist is masked off Romanized rows. We are deliberately modest about what the design’s named parts buy. On our own ablations the validation-optimised fusion lands within 0.0003 of plain averaging, and the script mask is overall-neutral (it helps only the Romanized slice), so the committee’s edge over its best single member comes from encoder diversity rather than from the weighting rule or the mask. A component ablation then reports the sensitivity of common imbalance additions under a fixed, single-backbone comparison.

(C4) A cross-source robustness study. Holding each Bangla-script source out in turn gives a source-transfer measurement and shows how far in-domain accuracy overstates deployment readiness. The fourth hold-out, on the sole Romanized source, moves source and script together and is scoped out of the analysis (Section 3.5).

(C5) A protocol-aligned comparison against the published base-paper comparator, plus a zero-shot LLM reference on the full test split.

(C6) Statistically grounded reporting and reproducibility materials. Headline F1, accuracy, and MCC figures, and every per-class F1, carry bootstrap 95% confidence intervals; the released artifact (Section 3.6) separates derived manifests and code from source-controlled text, so every table and figure can be regenerated from the cited source downloads.

The contribution is best read in that order: a benchmark, an evaluation protocol, and the empirical findings the two produce, with the proposed model as the fully specified reference system the protocol needs rather than an algorithmic novelty claim. The individual ingredients, pretrained encoders, FGM, weighted fusion, are all established; what is new is the resource, the measurement discipline around it, and what the measurements show. We scope that novelty carefully. Dual-script Bengali data and cross-dataset evaluation each have precedent—Ahmed et al. [2] already treat Bangla and Romanized text side by side, and transliteration-based resources exist—so we do not claim to be first on either axis. The claim is narrower: to be the first to harmonise four incompatible sources into one deduplicated five-class scheme and to run a systematic source-held-out transfer protocol over the result.

The rest of the paper is organised as follows. Section 2 reviews prior work; Section 3 describes the benchmark, its construction, and its governance; Section 4 formalises the proposed model on a single comment; Section 5 gives the experimental setup; Section 6 reports results; Section 7 discusses the findings and their limitations; and Section 8 concludes.

2. Related Work

2.1 Bengali Cyberbullying and Hate-Speech Detection

Early Bengali abuse detection leaned on classical machine learning over surface features. TF-IDF with logistic regression, SVMs, and tree ensembles was the standard toolkit [2–4], and it works until the abuse stops being lexical. Akhter et al. [3] pushed a hybrid machine-learning model for Bengali cyberbullying but stayed with a single Bangla-script source. Deep learning shifted the ceiling: BiLSTM and CNN-BiLSTM models with FastText or GloVe embeddings [5, 6] captured more context, though they too kept to single corpora. The move to transformers is recent and clear. Emon et al. [7] and Aurpa et al. [8] applied BERT-family encoders to Bangla abusive-comment detection; Karim et al. [9] added explainability to Bengali hate-speech classification; and Saifullah et al. [10] combined deep learning with transformer language models. The published comparator for this study, Hoque et al. [1], evaluates eight pretrained backbones and uses a top-three MLP stack on a 44,001-comment Facebook corpus. It is a strong single-source reference, but its protocol does not measure cross-source transfer and does not use FGM; our FGM result is a property of the proposed training recipe, not a claim about the comparator.

Datasets have grown alongside the methods. Romim et al. [11] released a Bengali hate-speech corpus with baseline evaluation; Ahmed et al. [12] built a Facebook harassment dataset with exploratory analysis; regional and multi-task resources such as BIDWESH [13] and LLM-based multi-task hate-speech corpora [14] broadened coverage into dialects and severity. Recent resources such as BanTH [15]—itself a large multi-label transliterated-Bangla dataset, and in fact one of the four sources we consolidate here—and BanCyB [16] also broaden the field, so our novelty claim is scoped to this consolidated benchmark and its evaluation protocol rather than to Bengali cyberbullying detection in general. Table 1 places BanglaCyberBench against representative prior resources on the axes that matter for this paper: number of sources, script coverage, taxonomy granularity, and whether cross-distribution transfer was ever measured.

Table 1. Representative Bengali abuse-detection resources against BanglaCyberBench. “Shift tested” asks whether the study measured transfer to an unseen source or script rather than only a random in-domain split.

Resource	Sources	Script	Classes	Shift tested	Size
Romim et al. [11]	1	Bangla	binary	no	30k
Ahmed et al. [2]	1	Bangla + Rom.	binary	no	10k
Karim et al. [9]	1	Bangla	multi	no	8k
Base paper [1]	1	Bangla	5	no	44k
BanglaCyberBench (ours)	4	Bangla + Rom.	5	yes	94k

2.2 Romanized Bangla and the Script Problem

Romanized Bangla is not a niche. A large fraction of Bengali social-media text is typed in the Latin alphabet, with no spelling standard, heavy code-mixing with English, and free vowel-dropping. Ahmed et al. [2] is one of the few studies to treat Bangla and Romanized Bangla side by side, and it does so with classical models on a modest scale. The pretrained-encoder landscape reflects the split: BanglaBERT [17] is an ELECTRA-style discriminator pretrained on 27.5 GB of Bangla-script text and is state of the art on Bangla-script understanding, while its sibling BanglishBERT is pretrained bilingually on Bangla and English and is the natural choice for Romanized input. MuRIL [18] adds transliterated Indian-language text, and XLM-R [19] is a broad multilingual baseline. A recurring, under-tested assumption is that one encoder can serve both scripts. Our robustness results question that assumption directly, and the script-aware design in Section 4 is our response to it.

2.3 Adversarial and Imbalance-Aware Fine-Tuning

Perturbing continuous word-embedding parameters during training is a cheap regulariser for small-data text classification. The Fast Gradient Method [20], a text adaptation of the image-domain idea of Goodfellow et al. [21], adds one adversarial forward-backward pass per batch. Complementary tricks target class imbalance and overfitting: focal loss [22] and class-balanced weighting [23] down-weight easy majority examples, multi-sample dropout [24] reduces head variance, R-Drop [25] enforces consistency between two dropout passes, weight averaging in the style of mean teachers [26] stabilises evaluation weights, and logit adjustment [27] corrects the long tail at inference. Our fixed single-backbone sensitivity run compares these choices descriptively; it does not establish a universal ordering. The proposed model uses CE+FGM as a transparent compromise between the measured gains and training cost.

3. The BanglaCyberBench Benchmark

BanglaCyberBench merges four public Bengali datasets into one deduplicated, dual-script, five-class corpus. Fig. 1 shows the construction pipeline end to end; this section walks through each stage and the decisions behind it.

What the task is, and is not. A word on the construct before the mechanics, because the name can mislead. BanglaCyberBench is a comment-level *harmful-language typing* task: given one isolated comment, assign the type of harm it carries—generic abuse, sexual or gender-based harassment, religion-directed abuse, or an explicit threat—or *none*. Cyberbullying as studied in the social-science literature usually also involves repetition, a power imbalance between perpetrator and target, a relationship or conversation history, and demonstrable harm to a victim. None of those signals survive in the four source corpora, which are single-comment collections without threads, user identities, or timestamps, so none of them can be modelled or measured here. We keep the “cyberbullying” label for continuity with the source datasets and the base paper’s lineage, but readers should take the actual object of study to be the harmful *content* of a comment, not the interactional phenomenon; a title of the form “harmonising multi-source Bengali harmful-language data” would describe the contribution just as well. This scoping also bounds the deployment claim: a system trained here flags harmful comments in isolation and is not, on its own, a cyberbullying detector in the fuller sense.

3.1 Sources and Schema Unification

The four sources differ in platform, script, and label format (Table 2). BANTH is Romanized Bangla with a binary column plus per-type indicator columns; FACEBOOK_44001 is Bangla script with a single label column and is the base paper’s source; MULTILABEL_12557 is Bangla with separate indicator columns; and BD_SHS is Bangla with one harmful-flag column and one type column. We rewrite all four into a common schema, `text · text_clean · label_binary · label_type · label5 · source · script · uid`, so that everything downstream reads one format. BANTH is the only Romanized source, which is why, later, source-held-out on BANTH and any cross-script analysis are closely related rather than independent.

Table 2. The four sources after deduplication. BANTH is the sole Romanized-Bangla source; the other three are Bangla script. Shares are of the 94,323-comment benchmark.

Source	Script	Origin	Samples	Raw label style
FACEBOOK_44001 [12]	Bangla	Mendeley Data	43,078	single label column
BANTH [15]	Romanized	public release	37,334	binary column + 0/1 type columns
MULTILABEL_12557 [28]	Bangla	Mendeley Data	8,882	separate 0/1 type columns
BD_SHS [29]	Bangla	ACL/repository	5,029	one harmful column + one type column
Total	—	—	94,323	—

3.2 Preprocessing

Bengali abuse leans on cues that aggressive normalisation destroys, so the cleaning is light and script-safe. We apply NFKC Unicode normalisation, strip zero-width and invisible characters, mask

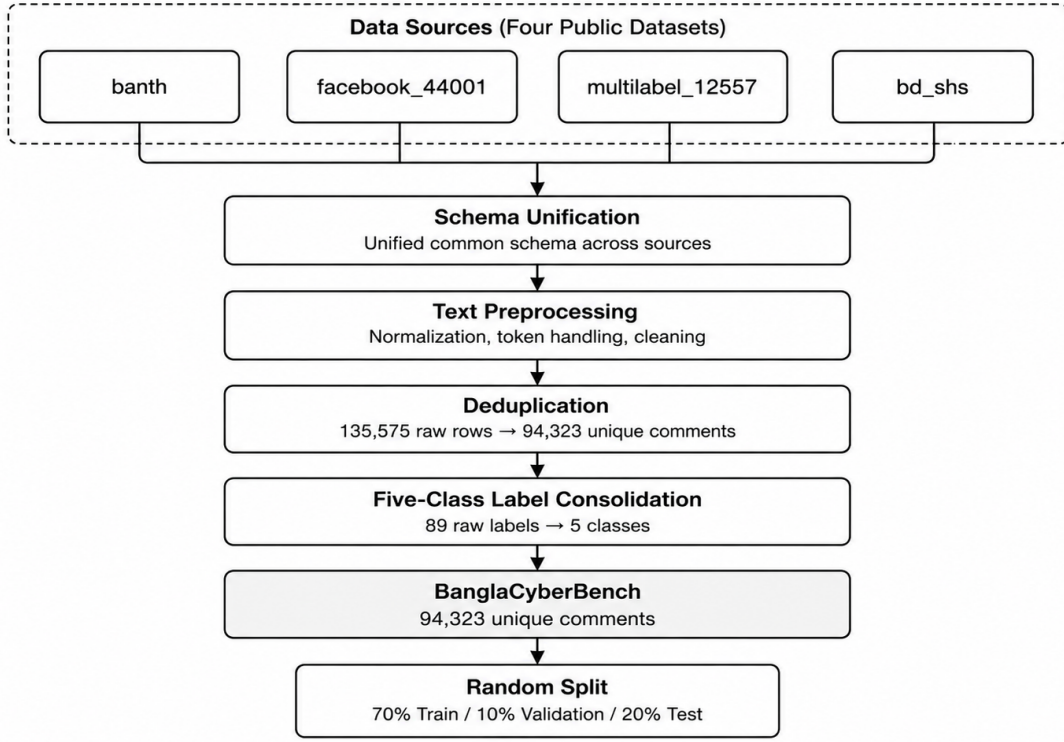


Figure 1. Benchmark construction pipeline. Four sources are unified into a common schema, lightly normalised, deduplicated on cleaned text (135,575 $\rightarrow$ 94,323), consolidated from 89 raw label strings into five classes by a priority rule, and split into a stratified random 70/10/20 partition plus source-held-out configurations. Every held-out configuration asserts a zero uid intersection between training and test.

URLs and mentions to [URL] and [USER], rewrite hashtags to [HASHTAG] while keeping the tag word, demojise emojis to bracketed text tokens, collapse runs of three or more identical characters down to two (so *baaaaje* and *baje* are not treated as unrelated), and normalise whitespace into `text_clean`. We deliberately do not lowercase-fold beyond Unicode rules, strip punctuation, or remove stopwords: emphatic repetition, punctuation, and emoji all carry abuse signal, and modern subword tokenisers handle the raw form well. Fig. 2 shows the composition of the resulting benchmark by source, script, and class.

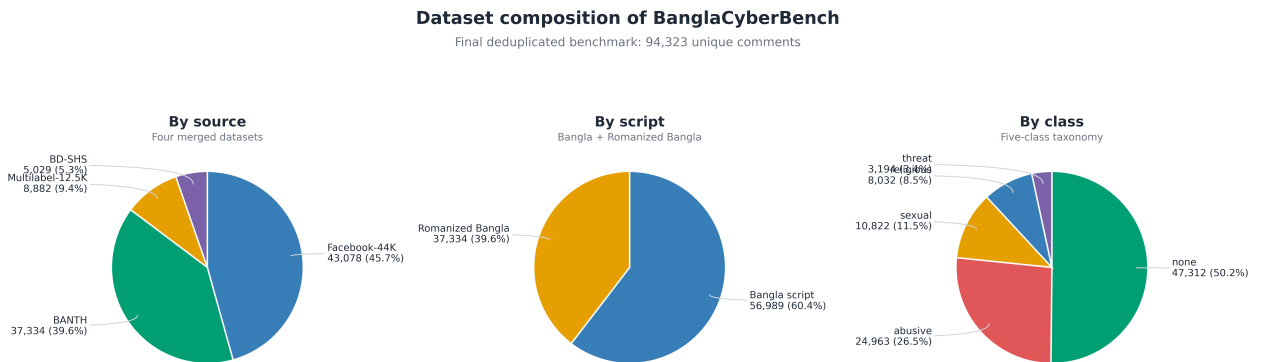


Figure 2. Composition of BanglaCyberBench after deduplication ($n = 94,323$). Left: the four merged sources. Middle: the near-even split between Bangla script (60.4%) and Romanized Bangla (39.6%). Right: the five-class taxonomy, dominated by *none* and *abusive* with a long tail down to *threat* at 3.4%.

3.3 Deduplication

Deduplication happens on `text_clean` and before any split is drawn. The merged corpus of 135,575 rows collapses to 94,323 unique comments, removing 41,252 duplicates. BANTH shrank the most because it carried many near-identical scrapes, which is why its post-deduplication share is smaller than its raw size would suggest. When duplicate rows disagree on their labels, the group resolves by the same

priority order as the taxonomy (Eq. (1)) and the binary flag is set if any duplicate was flagged harmful. For provenance, the retained row’s source and script are the source/script of the first row encountered in the deterministic merge order; this is the benchmark’s *primary-source* field, not an assertion that the comment originated there. An audit found 1,287 cleaned-text groups spanning multiple sources (3,308 raw rows); 661 groups (1,721 rows) had conflicting five-class labels and 271 groups (767 rows) had conflicting binary flags. These counts are reported so that the consolidation rule is auditable rather than hidden. The matching is exact on the cleaned text; fuzzy paraphrase detection is not applied (Section 7.4). A hard uid intersection assertion runs after every split to guarantee no retained comment appears in more than one partition.

3.4 The Five-Class Taxonomy

The four sources together use 89 distinct raw label strings, many of them comma- or underscore-compounded (a single comment might be tagged `sexual_threat` or `religious_abusive`). We split each raw string on `,` and `_`, map each piece to a bucket, and resolve compounds by a fixed priority:

$$\text{threat} > \text{sexual} > \text{religious} > \text{abusive} > \text{none}. \quad (1)$$

The order is deliberate. Explicit violence (*threat*) and identity-targeted abuse (*sexual*, *religious*) are the classes a moderator most needs surfaced, so they win over the generic *abusive* bucket when a comment carries several tags; *none* loses to everything. The priority order is a deterministic benchmark-construction policy motivated by moderation urgency, not a claim about a universal semantic hierarchy; a different governance context could justify a different order, and the released code makes re-consolidation under an alternative rule a one-line change. Consistency with the binary flag is enforced: a comment flagged non-abusive resolves to *none*, and one flagged abusive but resolving to *none* is placed in *abusive*. The resulting distribution (Table 3) is realistically imbalanced, with *none* at half the data and *threat* at 3.4%, a roughly 14.8:1 gap between the most and least frequent class. That imbalance is the core difficulty: Macro-F1 weights every class equally, so a model that leans on the majority class is punished exactly where moderation matters most.

Table 3. Class distribution of the five-class benchmark after deduplication ($n = 94,323$). Raw labels folded into each class are listed in the text.

Class	Samples	Share (%)
None	47,312	50.16
Abusive	24,963	26.47
Sexual	10,822	11.47
Religious	8,032	8.52
Threat	3,194	3.39
Total	94,323	100.00

The buckets are as follows. *None* folds in **none** and **not-bully**. *Abusive* is the catch-all for general offence: abusive/violence, troll, personal offence, body-shaming, origin-based slurs, slander, spam, political abuse, and miscellaneous. *Sexual* folds sexual and gender-based (misogynistic) harassment, *Religious* folds religion-directed abuse, and *Threat* folds explicit threats and calls to violence. Table 5 gives the full map. Two consistency rules run alongside it: an unrecognised component under a comment flagged abusive is sent to *abusive* rather than dropped, and the binary flag has the final say, so a comment flagged non-abusive resolves to *none* and one flagged abusive but resolving to *none* is moved to *abusive*.

One conceptual limitation of the scheme deserves stating plainly. The five labels do not sit on a single axis: *none* is a harmfulness decision, *threat* marks an act, *sexual* and *religious* are partly target-identity categories, and *abusive* is a residual content bucket. A few of the folds are genuinely arguable—gender-based abuse is grouped under *sexual*, and origin-based slurs land in the generic *abusive* bucket rather than an identity-targeted class—and a multi-label or hierarchical representation would capture such cases more faithfully than a flat forced choice. We keep the flat five-class rule for comparability with the source datasets and the base paper, and because the released code re-consolidates under an alternative rule in one line, but we do not claim the rule is the uniquely correct one; a sensitivity study across

consolidation policies (majority vote, source-specific, priority, conflict-exclusion) is the natural way to test that and is left to future work (Section 7.4).

To confirm the mechanical rule yields semantically sensible labels and not merely internally consistent ones, we measured inter-annotator agreement on 500 comments drawn with a fixed seed and stratified jointly by class and script. The stratification was proportional with a minimum cell size where possible; importantly, the consolidated Romanized subset contains no gold *threat* examples, so no Romanized-threat gold stratum is claimed. Two of the authors (S. Alam and N. Parvez) labelled the sample independently, blind to the consolidated label and to each other. They agreed substantially, Cohen’s κ [30] = 0.709 at 80.0% raw agreement (Table 4), and each matched the consolidated `label15` at $\kappa \approx 0.85$. Agreement was highest on *none* and thinnest on *threat*; Romanized items also showed greater ambiguity than Bangla items. Disagreements were left unadjudicated by design: the aim was to measure convergence on the consolidated scheme, not to build a corrected gold set. A larger study with more annotators, adjudication, and an interval estimate for κ is future work (Section 7.4).

Table 4. Inter-annotator agreement on a stratified sample of 500 comments, two authors labelling independently into the five classes.

Measure	Value	Interpretation
Cohen’s κ (annotators 1 vs. 2)	0.709	substantial
Raw agreement (1 vs. 2)	80.0%	—
κ (annotator 1 vs. consolidated label)	0.851	almost perfect
κ (annotator 2 vs. consolidated label)	0.852	almost perfect
Sample size	500	stratified

A nine-class variant that splits the catch-all further (adding *personal*, *political*, *gender*, *other*) is built only for the taxonomy ablation in Section 6.4, never for the headline, because those extra classes are too sparse to learn.

Table 5. Raw-label consolidation map. Each raw label string is lowercased and split on , and _; every component is sent to a bucket, and compounds are resolved by the priority order in Eq. (1).

Consolidated class	Raw components folded in
None	none, not bully, notbully
Abusive	abusive, abusive/violence, violence, troll, personal, personal offense, political, slander, spam, origin, body shaming, misc
Sexual	sexual, gender
Religious	religious, religion
Threat	threat, calltoviolence

3.5 Splits and Evaluation Protocol

The main split is a stratified random 70/10/20 partition on `label15`: 66,026 train, 9,432 validation, 18,865 test (Table 6). Stratification holds the five-class proportions fixed across all three partitions, so the rare *threat* and *religious* classes keep the same share in train, validation, and test rather than drifting by chance. Validation carries every fitted decision: early stopping, model selection within a run, the ensemble weights of Eq. (17), and any inference-time tuning. No hyperparameter, checkpoint, or fusion weight of the final model is chosen using test data, so the headline numbers of Section 6.1 are a clean held-out measurement. The random split measures generalisation to unseen comments from the same consolidated source mixture, not generalisation to a new platform or to future data; the source-held-out protocol below is our stand-in for the latter, and a truly external corpus remains future work (Section 7.4). We are explicit about one nuance: the ablations of Sections 6.4 and 6.5 report test-split scores, because they exist to tell the reader what each component is worth on the same yardstick as the headline. Their design conclusions rest on effects far larger than the gap between validation and test, and the proposed configuration was fixed before those tables were assembled, but a reader should treat the ablation tables as reported measurements rather than as the selection procedure itself.

On top of this we build *source-held-out* configurations over the three Bangla-script sources. In each, one Bangla source is held out as the test set and contributes all of its comments; the *other two Bangla-script*

sources are split into training and validation by a stratified 87.5/12.5 partition on `label15`. The sole Romanized source, BANTH, is left out of these three configurations entirely, on both sides: keeping it in training would place Romanized text on the training side of an otherwise Bangla-script test and quietly reintroduce the script shift the design is meant to hold fixed. This is why the training pool in Table 19 is the other two Bangla-script sources rather than all remaining data. One further scoping decision matters. A hold-out reads as a *source-shift* measurement only when the held-out source differs from the training pool in provenance but not in script; that holds for the three Bangla-script sources, and those three configurations form the source-shift study of Section 6.6. Holding out BANTH as a fourth configuration would instead strip every Romanized comment from training and test only on Romanized text, so its number would measure a compound source-plus-script shift attributable to neither factor alone; we omit it rather than report a number whose cause is unidentifiable. Disentangling the two needs a second Romanized source, the benchmark’s most valuable extension (Section 7.4). Every configuration re-runs the `uid` intersection assertion, so no retained comment appears in more than one partition—exact-match leakage is prevented by construction, with the residual near-duplicate caveat of Section 7.4 still applying.

One schema detail completes the protocol: the script tag $s(x)$ used throughout the paper is inherited from source provenance, BANTH comments are Romanized and the other three sources are Bangla script, rather than detected per comment. This is exact for the benchmark, whose sources are script-homogeneous. A deployed system facing free code-mixed input would need a per-comment script detector (a Unicode-block ratio is the standard choice), which we note rather than need here.

A scope note on the source-held-out design follows from the deduplication order (Section 3.3). Because dedup runs globally before splitting and a comment shared across sources is kept once under a single primary-source tag, the held-out test set is the benchmark’s set of rows *tagged* with the target source, not a from-scratch reconstruction of that source as originally collected. The 1,287 cleaned-text groups that span multiple sources (Section 3.3) therefore sit in training under whichever source they were first attributed to, even when their text also occurred in the held-out source. For the concordant-label majority of these groups the direction is clear: their shared text can only make the held-out source easier, so the transfer drops in Section 6.6 sit at or below a true from-scratch shift. We flag one caveat we cannot fully rule out. Among the 661 groups whose five-class labels *conflicted* across sources (Section 3.3), a training row tagged to one source may carry a label at odds with the held-out copy, and that need not make the hold-out easier. The net effect is therefore very likely conservative rather than provably so; the clean test is a from-scratch reconstruction that drops every target-source group from training, which we leave to future work (Section 7.4). The `uid` intersection assertion still guarantees that no single retained comment appears in two partitions.

Table 6. Stratified random 70/10/20 split of the benchmark. Stratification is on the five-class label.

Split	Samples	Share (%)
Train	66,026	70.00
Validation	9,432	10.00
Test	18,865	20.00

3.6 Data Governance and Release

A benchmark of abusive text needs its release policy stated as carefully as its construction. The four sources are cited at their published landing pages (Table 2); access and redistribution conditions remain those of the original providers. The public repository¹ contains the preprocessing, consolidation, split-construction, and evaluation code (as end-to-end notebooks), the raw-label consolidation map, the deterministic split-construction script and the seed that regenerates the UID-level manifests bit-for-bit, per-seed and fused-ensemble metric files, the fusion weights, the annotation gold sample and the zero-shot LLM baseline predictions, the per-seed ablation and robustness tables, all paper figures, and environment metadata. Raw comment text is redistributed only where the source licence permits it; otherwise users reconstruct the benchmark by downloading the cited sources and running the released pipeline, which is deterministic given the fixed seeds. The corpus contains abusive, sexual, religious,

¹Repository: <https://github.com/Sefayet-Alam/BanglaCyberBench>; the camera-ready snapshot is archived as the tagged, immutable release `v1.0.2`, and a Zenodo DOI will be added at camera-ready.

and threatening language, so the release carries a content warning, misuse guidance, and a documented route for source-specific removal requests.

Two release specifics go beyond a bare code drop. We ship a per-source data-licence file (`DATA_LICENSE.md`) that records, for each of the four sources, the terms under which it was obtained and whether its raw text may be redistributed, alongside pinned environment specifications and a SHA-256 manifest (`SHA256SUMS`) so a rebuild can be verified against exactly what we used. The artifact is versioned as a tagged, immutable GitHub release (<https://github.com/Sefayet-Alam/BanglaCyberBench/releases/tag/v1.0.3>), with a Zenodo DOI to be minted at camera-ready, and it ships a datasheet and a model card that state intended use, known gaps—the low Romanized score of Section 6.5 among them—and out-of-scope uses. Deletion requests propagate: the documented takedown route removes a flagged comment from the redistributed manifests and, where raw text cannot be redistributed, from the reconstruction index, so a removed item does not reappear on a later rebuild. One processing step also touches a third party. The zero-shot reference of Section 6.2 sent all 18,865 test comments to Gemini through the OpenRouter API; we treat that as a data transfer, assess it against the source licences (`DATA_LICENSE.md`) and the provider’s stated retention terms, and note that the benchmark reconstruction itself never requires it. We did not run a comprehensive PII audit, and we say so rather than imply one (Section 7.4).

4. Proposed Model

The proposed model is a script-aware ensemble of four fine-tuned transformer encoders, fused by a validation-optimised weighted-logit rule. This section traces the whole path a single comment takes through the system: tokenisation and embedding (Section 4.1), the encoder and classification head (Section 4.2), the training objective, cross-entropy plus FGM (Section 4.3), the script-aware masking that makes the ensemble aware of which alphabet a comment is written in (Section 4.4), and the weighted-logit fusion that produces the final label (Section 4.5). Fig. 3 shows the assembled system. Table 7 fixes notation.

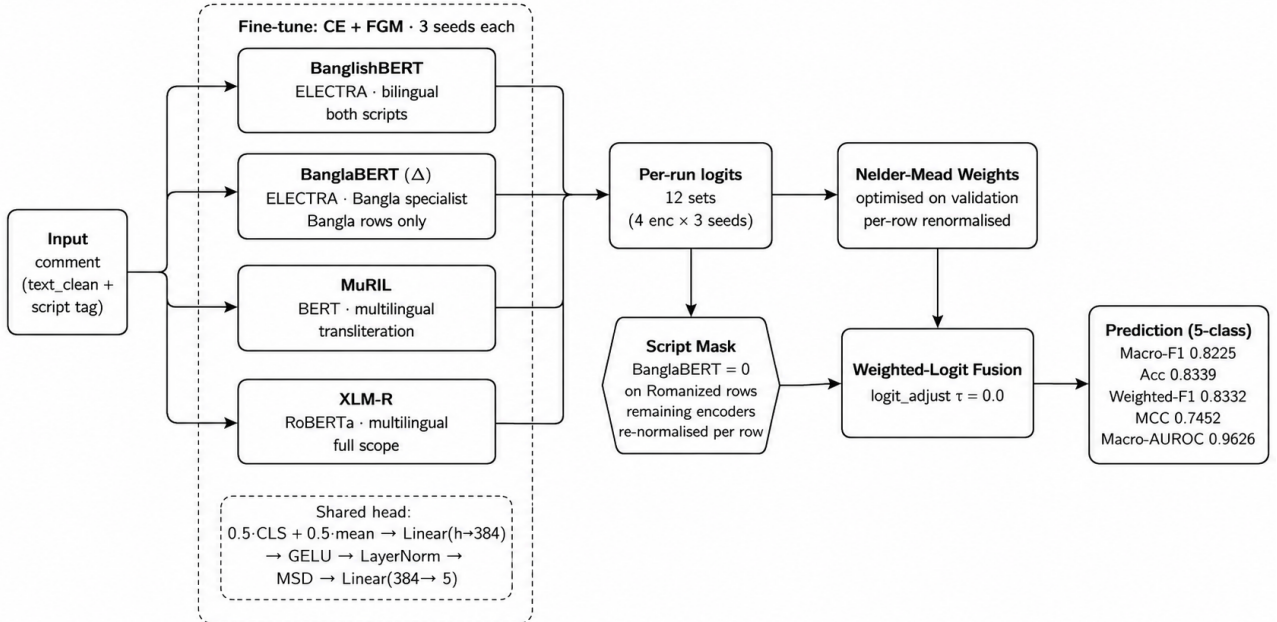


Figure 3. The proposed script-aware ensemble. A comment is routed by script. All four encoders process Bangla-script comments; on Romanized comments the Bangla-script specialist BanglaBERT emits neutral logits and receives zero fusion weight, so the remaining encoders decide the label. Each encoder is fine-tuned with cross-entropy plus FGM across three seeds, and the twelve resulting logit vectors are fused by validation-optimised weights.

Table 7. Notation used in Section 4.

Symbol	Meaning
x, y	Input comment (token sequence) and its ground-truth class
T, C	Sequence length (≤ 128 tokens) and number of classes ($C = 5$)
$s(x) \in \{\text{BN}, \text{ROM}\}$	Script of comment x : Bangla or Romanized
$\mathbf{e} \in \mathbb{R}^{T \times d}$	Input-embedding matrix (token + position), $d = 768$
H, d_k, d_{ff}	Attention heads, per-head width $d_k = d/H$, and FFN inner width (3072)
$\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V$	Query, key, value projections of head i
θ_m, ϕ_m	Encoder and head parameters of backbone m
$\mathbf{h}_t \in \mathbb{R}^d$	Contextual hidden state at position t ; $\mathbf{h}_{[\text{CLS}]} \equiv \mathbf{h}_0$
$\mathbf{z}_m \in \mathbb{R}^C$	Logit vector from backbone m
$\hat{p}(y x) \in \Delta^{C-1}$	Predicted class-probability vector
$\mathbf{E}_m, \mathbf{r}$	Token-embedding parameter of member m and its temporary FGM perturbation
ϵ	FGM perturbation budget
$w_m \geq 0$	Fusion weight of ensemble member m , $\sum_m w_m = 1$
$\mathcal{M}, \mathcal{M}(x)$	Full member set (12) and the active members for comment x

4.1 From a Comment to Contextual Embeddings

Take a single Romanized comment, $x = \text{"tui akta faltu manush"}$ (roughly, “you are a worthless person”). A backbone’s subword tokeniser turns it into a sequence of T token ids ($w_1, \dots, w_T$) with a leading [CLS] and trailing [SEP]. Each id is mapped to a d -dimensional vector by a token-embedding lookup $\mathbf{E}_{\text{tok}} \in \mathbb{R}^{V \times d}$ and added to a position embedding, so the input row for position t is

$$\mathbf{e}_t = \mathbf{E}_{\text{tok}}[w_t] + \mathbf{E}_{\text{pos}}[t], \quad \mathbf{e} = (\mathbf{e}_1, \dots, \mathbf{e}_T) \in \mathbb{R}^{T \times d}, \quad (2)$$

with $d = 768$. The encoder is a stack of $L = 12$ transformer blocks [31]. Each block first contextualises the sequence $\mathbf{H} \in \mathbb{R}^{T \times d}$ with multi-head self-attention. For head $i \in \{1, \dots, H\}$ the block projects $\mathbf{H}$ into queries, keys, and values, then applies scaled dot-product attention,

$$\mathbf{Q}_i = \mathbf{H}\mathbf{W}_i^Q, \quad \mathbf{K}_i = \mathbf{H}\mathbf{W}_i^K, \quad \mathbf{V}_i = \mathbf{H}\mathbf{W}_i^V, \quad \text{head}_i = \text{softmax}\left(\frac{\mathbf{Q}_i\mathbf{K}_i^\top}{\sqrt{d_k}} + \mathbf{M}\right)\mathbf{V}_i, \quad (3)$$

where $\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V \in \mathbb{R}^{d \times d_k}$, $d_k = d/H$, and $\mathbf{M}$ is an additive mask sending padded positions to $-\infty$ so they receive zero attention weight. The row-wise softmax produces an attention matrix $\mathbf{A}_i \in \mathbb{R}^{T \times T}$ whose entry $A_{i,ts}$ is the weight token t places on token s ; this is the mechanism that binds *faltu* (“worthless”) to *manush* (“person”), since the insult lives in their pairing rather than either token alone. The H heads are concatenated and mixed by an output projection $\mathbf{W}^O \in \mathbb{R}^{d \times d}$,

$$\text{MHA}(\mathbf{H}) = \text{Concat}(\text{head}_1, \dots, \text{head}_H)\mathbf{W}^O. \quad (4)$$

A position-wise feed-forward network then lifts each token to an inner width $d_{\text{ff}} = 3072$ and back,

$$\text{FFN}(\mathbf{x}) = \text{GELU}(\mathbf{x}\mathbf{W}_1^{\text{ff}} + \mathbf{b}_1^{\text{ff}})\mathbf{W}_2^{\text{ff}} + \mathbf{b}_2^{\text{ff}}, \quad \mathbf{W}_1^{\text{ff}} \in \mathbb{R}^{d \times d_{\text{ff}}}, \mathbf{W}_2^{\text{ff}} \in \mathbb{R}^{d_{\text{ff}} \times d}, \quad (5)$$

and each sub-layer is wrapped in a residual connection and layer normalisation,

$$\mathbf{H}' = \text{LN}(\mathbf{H} + \text{MHA}(\mathbf{H})), \quad \mathbf{H}'' = \text{LN}(\mathbf{H}' + \text{FFN}(\mathbf{H}')), \quad (6)$$

where $\text{LN}(\mathbf{x}) = \gamma \odot (\mathbf{x} - \mu) / \sqrt{\sigma^2 + \epsilon} + \beta$ standardises each token vector over its d features using the per-token mean μ and variance σ^2 , with learned scale γ and shift β . Stacking the twelve blocks on top of $\mathbf{e}$ yields contextual hidden states $(\mathbf{h}_0, \dots, \mathbf{h}_{T-1})$, collecting the parameters of all blocks, embeddings, and projections into the encoder set θ_m .

Two of our backbones, BanglaBERT and BanglishBERT [17], are pretrained as ELECTRA discriminators [32] rather than masked language models: a small generator replaces some tokens and the discriminator learns to flag which ones were replaced, an objective that yields strong Bengali representations from 27.5 GB of text. MuRIL and XLM-R are standard masked-LM encoders. For our purposes

the pretraining objective matters only through the representations it produces; the fine-tuning path below is identical across all four.

4.2 Pooling and Classification Head

Each backbone m carries the same lightweight head. We pool the sequence by blending the [CLS] vector with a mean over token states, which is steadier than [CLS] alone on short, noisy comments:

$$\mathbf{h}_{\text{pool}} = \frac{1}{2} \mathbf{h}_{[\text{CLS}]} + \frac{1}{2} \cdot \frac{\sum_{t=1}^T m_t \mathbf{h}_t}{\sum_{t=1}^T m_t}, \quad (7)$$

where $m_t \in \{0, 1\}$ is the attention mask that zeroes out padding. The pooled vector passes through a small projection with GELU and LayerNorm, then a linear layer to C logits:

$$\mathbf{u} = \text{LN}(\text{GELU}(\mathbf{W}_1 \mathbf{h}_{\text{pool}} + \mathbf{b}_1)) \in \mathbb{R}^{384}, \quad \mathbf{z}_m = \mathbf{W}_2 \mathbf{u} + \mathbf{b}_2 \in \mathbb{R}^C, \quad (8)$$

with $\mathbf{W}_1 \in \mathbb{R}^{384 \times d}$ and $\mathbf{W}_2 \in \mathbb{R}^{C \times 384}$. The class probabilities are $\hat{p}(y | x) = \text{softmax}(\mathbf{z}_m)$. For our example comment, a well-trained encoder puts most of the softmax mass on *abusive*.

4.3 Training Objective: Cross-Entropy with FGM

Each backbone is fine-tuned end-to-end (both θ_m and ϕ_m) with AdamW [33] under a mild label-smoothed cross-entropy. With smoothing $s = 0.03$ the soft target is $\tilde{y}_c = (1 - s)\mathbb{I}[c = y] + s/C$, and

$$\mathcal{L}_{\text{CE}}(\theta_m, \phi_m) = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C \tilde{y}_{i,c} \log \hat{p}(y = c | x_i). \quad (9)$$

Cross-entropy alone leaves the decision boundary brittle in directions the training set does not cover, which for user-generated Bengali means small spelling changes can flip a prediction. Adversarial training addresses this by asking, for each batch, what small perturbation of the input embeddings would most increase the loss, and then training against it. Formally it is a min-max problem,

$$\min_{\theta_m, \phi_m} \mathbb{E}_{(x,y)} \left[\max_{\|\mathbf{r}\|_F \leq \epsilon} \mathcal{L}_{\text{CE}}(\mathbf{E}_m + \mathbf{r}; x, y) \right]. \quad (10)$$

FGM [20] is implemented here as a temporary perturbation of the trainable token-embedding parameter, rather than as a persistent change to the input ids. Let $\mathbf{E}_m$ denote the token-embedding parameter of backbone m and let $\mathbf{g}_E = \nabla_{\mathbf{E}_m} \mathcal{L}_{\text{CE}}(\mathbf{E}_m, y)$ be its clean-pass gradient. A first-order Taylor expansion gives $\mathcal{L}_{\text{CE}}(\mathbf{E}_m + \mathbf{r}, y) \approx \mathcal{L}_{\text{CE}}(\mathbf{E}_m, y) + \langle \mathbf{r}, \mathbf{g}_E \rangle_F$, so the inner problem under a Frobenius-norm budget is maximised by

$$\boxed{\mathbf{r}_{\text{adv}} = \epsilon \frac{\mathbf{g}_E}{\|\mathbf{g}_E\|_F}} \quad (11)$$

a single ascent step on the embedding-parameter ball. The perturbation is applied in-place to the embedding weight for the adversarial forward pass and then restored, so the optimiser never treats it as a new parameter. The per-batch objective sums the clean and adversarial forward passes,

$$\mathcal{L}(\theta_m, \phi_m) = \mathcal{L}_{\text{CE}}(\mathbf{E}_m; x, y) + \mathcal{L}_{\text{CE}}(\mathbf{E}_m + \mathbf{r}_{\text{adv}}; x, y), \quad (12)$$

and both passes accumulate gradients into the same parameters before a single optimiser step (Algorithm 1). On “*tui akta faltu manush*” the affected rows of the embedding table are nudged in the loss-increasing direction. FGM costs one extra forward-backward pass per batch and exposes a single hyperparameter ϵ , fixed at 1.0.

Parameter updates. Both the encoder θ_m and the head ϕ_m are trained end to end. Writing $\Theta = (\theta_m, \phi_m)$ for all trainable parameters, the gradient from Eq. (12) flows back through the head (Eqs. (7) and (8)) and then through every transformer block (Eqs. (3), (4) and (6)) by the chain rule; for a single weight matrix $\mathbf{W}$ in block ℓ this is $\partial \mathcal{L} / \partial \mathbf{W} = \sum_t (\partial \mathcal{L} / \partial \mathbf{H}_t^{(\ell)}) (\partial \mathbf{H}_t^{(\ell)} / \partial \mathbf{W})$, with the attention softmax and GELU contributing their Jacobians along the path. The accumulated gradient $\mathbf{g}_\Theta = \nabla_\Theta \mathcal{L}$ drives an AdamW update [33], which maintains bias-corrected first- and second-moment estimates $\hat{\mathbf{m}}_t, \hat{\mathbf{v}}_t$ and

decouples weight decay from the gradient step,

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \mathbf{g}_\Theta, \quad \mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2) \mathbf{g}_\Theta^2, \quad (13)$$

$$\hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1 - \beta_1^t}, \quad \hat{\mathbf{v}}_t = \frac{\mathbf{v}_t}{1 - \beta_2^t},$$

$$\Theta_t = \Theta_{t-1} - \eta \left(\frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \varepsilon} + \lambda \Theta_{t-1} \right), \quad (14)$$

with $\beta_1 = 0.9$, $\beta_2 = 0.999$, decoupled weight decay λ , and a learning rate η warmed up linearly and then decayed. The encoder and head use separate learning rates (Table 8); gradients are clipped in ℓ_2 norm before Eq. (14) to bound the step. Why FGM and nothing heavier is the subject of the ablation in Section 6.4.

Algorithm 1 Cross-entropy + FGM fine-tuning of one backbone (single mini-batch).

Require: Batch (x, y) , encoder θ_m , head ϕ_m , budget ϵ

- 1: Embed: $\mathbf{e} \leftarrow \text{Embed}(x)$
 - 2: **Clean pass:** $\mathcal{L}_{\text{clean}} \leftarrow \mathcal{L}_{\text{CE}}(\mathbf{e}, y)$; **backward** (accumulate $\nabla_{\theta_m, \phi_m}$)
 - 3: Read $\mathbf{g}_E = \nabla_{\mathbf{E}_m} \mathcal{L}_{\text{clean}}$; set $\mathbf{r}_{\text{adv}} \leftarrow \epsilon \mathbf{g}_E / \|\mathbf{g}_E\|_F$ $\triangleright$ Eq. (11)
 - 4: Temporarily set $\mathbf{E}_m \leftarrow \mathbf{E}_m + \mathbf{r}_{\text{adv}}$
 - 5: **FGM pass:** $\mathcal{L}_{\text{adv}} \leftarrow \mathcal{L}_{\text{CE}}(x, y)$; **backward** (accumulate); **restore** $\mathbf{E}_m$
 - 6: Clip gradients; **optimiser step:** $\theta_m, \phi_m \leftarrow$ AdamW update with configured encoder/head learning-rate groups and decoupled decay
-

4.4 Script-Aware Specialisation

The four backbones do not have equal competence on both scripts, and pretending they do wastes the strongest one. BanglaBERT is pretrained on Bangla-script text only; feeding it Romanized Bangla degrades its representations and, worse, drags the ensemble down on exactly the script it cannot read. We therefore make BanglaBERT a Bangla-script specialist. It trains, validates, and is evaluated only on Bangla-script rows. On a Romanized comment it emits a neutral zero logit vector and receives zero fusion weight, so it never votes on text it was not built for. BanglishBERT, pretrained bilingually, is the workhorse for Romanized input; MuRIL and XLM-R stay full-scope over both scripts. This gives the ensemble a clean specialist-plus-generalists structure: for a Bangla comment all four members are active, and for a Romanized comment the three script-capable members decide.

Concretely, define an activation indicator for member m on comment x :

$$a_m(x) = \begin{cases} 0 & \text{if } m \text{ is a BanglaBERT member and } s(x) = \text{ROM}, \\ 1 & \text{otherwise.} \end{cases} \quad (15)$$

The active set is $\mathcal{M}(x) = \{m \in \mathcal{M} : a_m(x) = 1\}$. For our Romanized example, the three BanglaBERT seed members drop out and the nine remaining members carry the decision.

One caveat about $s(x)$ belongs here rather than only in the protocol section. On this benchmark the script tag is read off source provenance—BANTH is Romanized, the other three sources are Bangla script—and is not detected per comment (Section 3.5). Because every source is script-homogeneous, script and source are perfectly correlated in our data, so the routing in Eq. (15) is exact but is effectively source-derived oracle metadata. A deployment facing free code-mixed input would replace $s(x)$ with a per-comment script detector (a Unicode-block ratio is the usual choice) and inherit its errors; the numbers reported here do not include that source of routing error.

4.5 Weighted-Logit Fusion

The ensemble has $|\mathcal{M}| = 12$ members: four backbones, each fine-tuned under three seeds (42, 123, 456). Fusion is a script-masked weighted average of logits, renormalised per comment over the active members so that dropping BanglaBERT on Romanized rows does not shrink the total weight:

$$\mathbf{z}_{\text{ens}}(x) = \sum_{m \in \mathcal{M}(x)} \frac{w_m}{\sum_{m' \in \mathcal{M}(x)} w_{m'}} \mathbf{z}_m(x), \quad \hat{y}(x) = \arg \max_c [\mathbf{z}_{\text{ens}}(x)]_c. \quad (16)$$

The weights are the solution of a constrained maximisation of validation Macro-F1 over the probability simplex,

$$\mathbf{w}^* = \arg \max_{\mathbf{w} \in \Delta^{11}} \text{F1}_{\text{macro}}^{\text{val}}(\mathbf{w}), \quad \Delta^{11} = \left\{ \mathbf{w} \in \mathbb{R}^{12} : w_m \geq 0, \sum_m w_m = 1 \right\}, \quad (17)$$

where $\text{F1}_{\text{macro}}^{\text{val}}(\mathbf{w})$ is the Macro-F1 obtained on the validation split when Eq. (16) is decoded with weights $\mathbf{w}$. This objective is piecewise constant in $\mathbf{w}$: its value changes only when some comment’s argmax flips, so the gradient is zero almost everywhere and undefined at the flip boundaries. Gradient methods are therefore useless here, which is why we solve Eq. (17) with the derivative-free Nelder–Mead simplex search [34]. The algorithm maintains a simplex of thirteen candidate weight vectors in $\mathbb{R}^{12}$ and, at each iteration, replaces the worst vertex using the standard reflection, expansion, contraction, and shrink moves (coefficients $\alpha = 1$, $\gamma = 2$, $\rho = \frac{1}{2}$, $\sigma = \frac{1}{2}$); candidates that leave the simplex are clipped at zero and renormalised, which projects them back onto Δ^{11} . The search starts from the uniform vector $w_m = \frac{1}{12}$, so the optimiser has to earn any departure from plain averaging. Each objective evaluation is cheap because the twelve members’ validation logits are computed once and cached; evaluating a candidate $\mathbf{w}$ is then a weighted sum, an argmax, and one Macro-F1 computation over the 9,432 validation rows, so the whole search costs seconds. The weights are fitted once, frozen before the 20% test set is ever seen, and reported in Section 6.8, so the test numbers stay clean. An optional inference-time logit adjustment for the long tail [27] is available but tuned to zero strength on this benchmark, so the fused logits above are the final decision.

4.6 A Worked Example on Two Scripts

Following one comment through the system in each script makes the routing concrete. Fig. 4 traces both cases end to end.

Take the Romanized comment $x_1 = \text{“tui akta faltu manush”}$. Its script is $s(x_1) = \text{ROM}$, so Eq. (15) switches the three BanglaBERT seed members off and $\mathcal{M}(x_1)$ holds nine members. The tokeniser produces [CLS] tui akta fal ##tu manush [SEP] and the embedding layer gives $\mathbf{e} \in \mathbb{R}^{T \times 768}$ (Eqs. (2) and (3)). Each active encoder pools through Eqs. (7) and (8) into a logit vector; a representative one places its mass at roughly [ab 0.61, no 0.20, se 0.11, re 0.04, th 0.04]. During training, FGM temporarily perturbs the relevant rows of the embedding parameter along the loss gradient (Eq. (11)); fusion then renormalises the nine active weights and averages the logits (Eq. (16)), whose argmax is *abusive*.

For the Bangla-script comment $x_2 = \text{tui akta baje lok}$ (“you are a bad person”), $s(x_2) = \text{BN}$, every $a_m(x_2) = 1$, and all twelve members vote, the specialist included. Pooling, head, and fusion are unchanged and the argmax is again *abusive*. The two traces differ in exactly one place, which members are active, and that single difference is the whole of the script-aware design.

5. Experimental Setup

5.1 Backbones and Training Configuration

The four encoders are BanglishBERT (`csebuetnlp/banglishbert`, bilingual), BanglaBERT (`csebuetnlp/banglabert`, Bangla-script specialist), MuRIL (`google/muril-base-cased`), and XLM-R (`xlm-roberta-base`). All share the pooling and head of Section 4.2. Table 8 lists the training configuration, held fixed across backbones so that any difference between them reflects the encoder, not incidental tuning. The learning rates are uniform across encoder layers, with decoupled AdamW weight decay applied to the configured parameter groups; this is distinct from layer-wise learning-rate decay. The effective batch size is 32. All runs use fp16 mixed precision on a single NVIDIA A6000 (48 GB) with 32 GB system RAM; software is PyTorch [35] with HuggingFace Transformers [36] and scikit-learn [37]. Seeds are fixed at {42, 123, 456}.

5.2 Evaluation Metrics

The primary metric is Macro-F1, reported with Weighted-F1 alongside. For C classes,

$$\text{F1}(c) = \frac{2 \text{Prec}(c) \text{Rec}(c)}{\text{Prec}(c) + \text{Rec}(c)}, \quad \text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C \text{F1}(c). \quad (18)$$

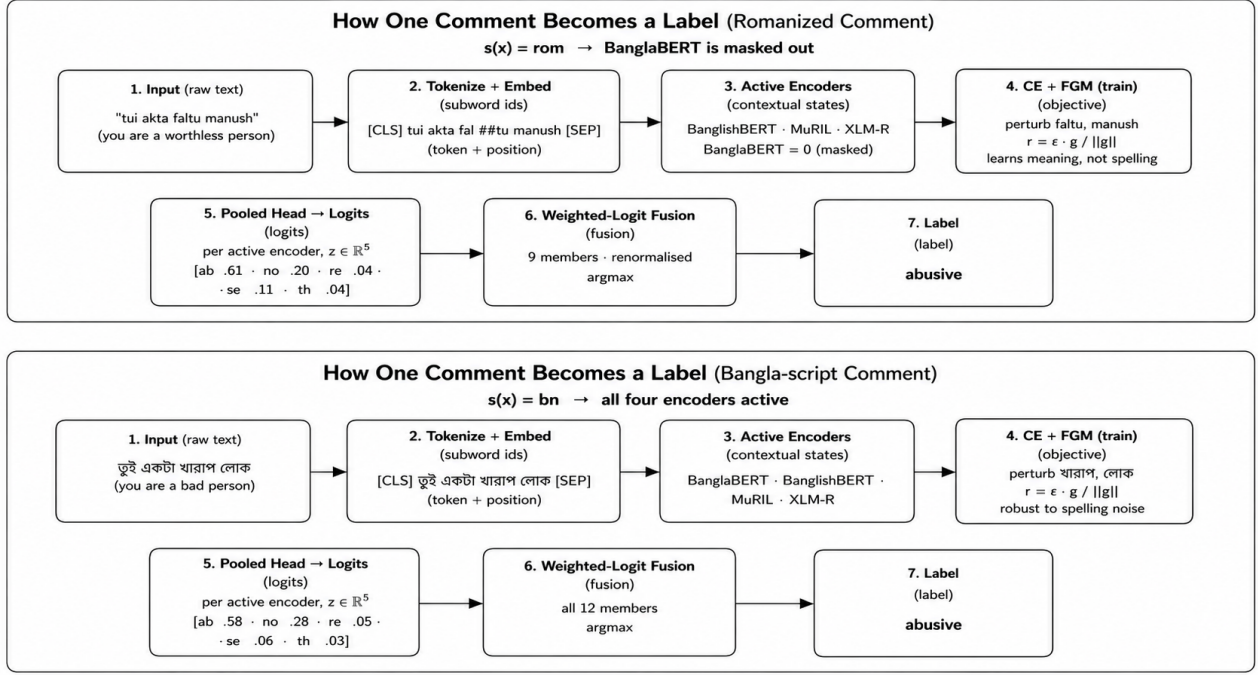


Figure 4. One comment traced through the system in each script. Top: a Romanized comment, where the Bangla-script specialist is masked out and nine members decide. Bottom: a Bangla-script comment, where all twelve members vote. Every other stage, tokenisation, embedding, encoding, the CE+FGM objective, pooling, and weighted-logit fusion, is identical across the two scripts.

Table 8. Training configuration for the four backbones. Held constant across encoders.

Hyperparameter	Value	Notes
Optimiser	AdamW	$\beta_1 = 0.9, \beta_2 = 0.999$
Encoder / head learning rate	$2 \times 10^{-5} / 8 \times 10^{-5}$	uniform across layers, no layer-wise decay
AdamW weight decay λ	0.01	decoupled, Eq. (14); excludes bias and LayerNorm
LR schedule	linear	10% warm-up, then linear decay
Gradient clipping	1.0	global ℓ_2 norm
Model selection	best validation Macro-F1	checkpoint restored before test
Max sequence length	128	covers nearly the whole length distribution
Batch size (effective)	32	physical $32 \times$ accumulation 1
Epochs	up to 8	early stopping, patience 3
Label smoothing	0.03	mild
Dropout / head hidden	0.25 / 384	—
FGM budget ϵ	1.0	embedding-space perturbation
Precision	fp16	A6000 48 GB
Seeds	42, 123, 456	three per backbone

Macro-F1 weights every class equally, so the rare *threat* and *religious* classes are not hidden by the dominant *none* class; Weighted-F1, which weights by support, is reported for continuity with prior work. We also report Accuracy and the Matthews correlation coefficient [38], which is informative under imbalance because it uses every cell of the confusion matrix $\mathbf{K} \in \mathbb{N}^{C \times C}$. In its multiclass form,

$$\text{MCC} = \frac{\sum_c K_{cc} \cdot N - \sum_c \hat{n}_c n_c}{\sqrt{(N^2 - \sum_c \hat{n}_c^2)(N^2 - \sum_c n_c^2)}}, \quad (19)$$

where N is the number of test comments, $n_c = \sum_j K_{cj}$ the true count of class c , and $\hat{n}_c = \sum_i K_{ic}$ its predicted count; the coefficient is 1 for perfect prediction, 0 at chance level, and is not inflated by majority-class dominance the way accuracy is. Finally we report one-vs-rest Macro-AUROC, the mean over classes of the area under the ROC curve treating each class against the rest, which measures the quality of the model’s ranking independently of any decision threshold.

5.3 Statistical Reporting

Because the 20% test set is a single fixed sample, we quantify its uncertainty with a percentile bootstrap [39]. For a metric T we draw $B = 2,000$ resamples of the 18,865 test predictions with replacement, recompute T on each, and report the interval between the 2.5th and 97.5th percentiles:

$$\text{CI}_{95\%} = [\hat{T}^{(\lceil 0.025B \rceil)}, \hat{T}^{(\lceil 0.975B \rceil)}]. \quad (20)$$

The resampling seed is fixed at 42. These intervals are conditional on one fixed test split and one trained final ensemble: they capture test-set sampling noise but not training-seed, source, annotation, or split-construction variance, so they should be read as lower bounds on total uncertainty rather than a full error budget. We report interval estimates rather than paired significance tests against the classical baselines, since those baselines are reference points rather than the comparison of interest.

5.4 Computational Cost

The whole system fits on a single 48 GB GPU, which matters in a low-resource setting where clusters are not a given. Each backbone carries 110–135M parameters, so the twelve fine-tuning runs are the cost, not the fusion: the Nelder–Mead search runs over twelve scalars on cached validation logits and takes seconds. On the A6000 a backbone fine-tunes in 35–50 minutes at length 128 and batch 32, so the full ensemble reproduces in a few GPU-hours. FGM doubles a run’s per-batch backward cost, but only in training; at inference the model is twelve forward passes (nine on Romanized input) plus a weighted average. We do not call the system lightweight. Twelve encoders is a committee, and a deployment that cannot afford one has a fallback in our own tables: the strongest full-scope single model (MuRIL, seed 123) reaches 0.8084 Macro-F1 on the same split (Table 11), giving up 0.0141 for a twelvefold cut in inference cost. That fallback must be full-scope, since the Bangla-script specialist abstains on Romanized input by design. Distilling the committee into one student is the natural production path and is left to future work.

5.5 Component Ablation Design

The proposed recipe fine-tunes with cross-entropy plus FGM. Whether either piece earns its place, and whether anything beyond the pair does, is the question the ablation settles rather than assumes. We anchor on BanglaBERT and take plain cross-entropy as the reference. From there we add one component at a time, first FGM, then each of the standard imbalance and regularisation tools that the usual playbook stacks on top, and measure the change in Macro-F1. Every configuration is run over three seeds, so each number is a mean with a standard deviation and the small deltas can be judged against seed noise instead of at face value (Section 6.4). The candidate components are defined here so their contributions are interpretable.

Class-balanced focal loss. Focal loss [22] down-weights easy, confident examples so the optimiser attends to hard ones, and class-balanced weighting [23] scales each class by the inverse of its effective sample count. With focusing parameter γ and per-class weight α_c ,

$$\mathcal{L}_{\text{focal}} = -\frac{1}{N} \sum_{i=1}^N \alpha_{y_i} (1 - \hat{p}_{i,y_i})^\gamma \log \hat{p}_{i,y_i}, \quad \alpha_c \propto \frac{1 - \beta}{1 - \beta^{n_c}}, \quad (21)$$

where n_c is the count of class c and $\beta \rightarrow 1$ recovers inverse-frequency weighting. We use $\gamma = 2$ and $\beta = 0.999$.

Balanced sampler. Instead of reweighting the loss, this reshapes the batch: each example is sampled with probability proportional to $n_{y_i}^{-1/2}$ (square-root inverse frequency), so minority classes appear more often during training.

Multi-sample dropout. The head is evaluated under K independent dropout masks and the losses are averaged, which reduces the variance of the head update at no inference cost [24]:

$$\mathcal{L}_{\text{MSD}} = \frac{1}{K} \sum_{k=1}^K \mathcal{L}_{\text{CE}}(\text{head}(\text{dropout}_k(\mathbf{h}_{\text{pool}}))). \quad (22)$$

R-Drop. Two independent dropout passes over the same input give two distributions $\hat{p}^{(1)}, \hat{p}^{(2)}$; R-Drop [25] adds their symmetric KL divergence to the loss, pushing the two passes to agree and so regularising against dropout noise:

$$\mathcal{L}_{\text{RDrop}} = \mathcal{L}_{\text{CE}} + \frac{\lambda}{2} [\text{KL}(\hat{p}^{(1)} \parallel \hat{p}^{(2)}) + \text{KL}(\hat{p}^{(2)} \parallel \hat{p}^{(1)})], \quad (23)$$

with $\lambda = 0.5$.

Exponential moving average (EMA). A shadow copy of the weights is maintained as an exponential moving average, $\bar{\theta} \leftarrow \mu \bar{\theta} + (1 - \mu) \theta$ with $\mu = 0.999$, and used for evaluation. It is kept only when it beats the raw weights on validation, so it can never hurt.

Each component is a configuration toggle, which makes the ablation a direct read on what earns its place. The full-stack configuration turns all of them on at once.

6. Results and Analysis

We report in-domain benchmark results with bootstrap confidence intervals (Section 6.1), a zero-shot LLM reference (Section 6.2), a per-class and error breakdown (Section 6.3), the ablations that fix the training recipe and taxonomy (Section 6.4), the effect of the script mask on Romanized-script Macro-F1 (Section 6.5), cross-source robustness (Section 6.6), the same-protocol comparison against the base paper (Section 6.7), and the learned ensemble weights (Section 6.8).

6.1 In-Domain Benchmark

On the official 20% test split ($n = 18,865$) the proposed model reaches Macro-F1 0.8225, Weighted-F1 0.8332, Accuracy 0.8339, MCC 0.7452, and Macro-AUROC 0.9626 (Table 9). The strongest non-transformer reference, a word+character TF-IDF linear SVM, reaches 0.7674 Macro-F1; the proposed model is 5.5 Macro-F1 points above it. The comparison is not data-budget matched: the classical baselines are fitted on train+validation (80%), needing no checkpoint selection, whereas the transformers optimise on train (70%) and use validation for early stopping. If anything this favours the classical reference, so the gap is not an artefact of giving the transformers more data. A deterministic percentile bootstrap over the retained test predictions gives the intervals in Table 10; these quantify uncertainty for the proposed model and are not a paired significance test against the reference, whose per-example predictions were not retained. We report the classical and zero-shot systems as reference points rather than the comparison of interest, so no paired test is claimed for them; the zero-shot baseline’s per-comment predictions and the fused model’s per-seed and ensemble outputs are in the released artifact, so the comparison can be extended by third parties. Fig. 5a shows the comparison, and Fig. 5b shows the intervals.

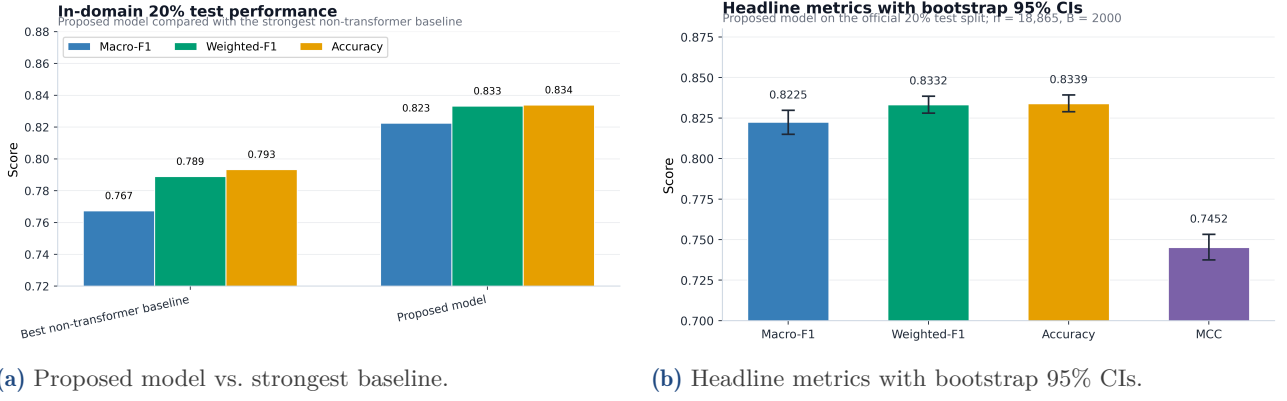
Table 9. Main results on the official 20% in-domain test split ($n = 18,865$). The reference is the strongest non-transformer baseline (TF-IDF word+char linear SVM).

System	Macro-F1	Weighted-F1	Accuracy	MCC	Macro-AUROC
Best non-transformer baseline	0.7674	0.7889	0.7933	0.6788	0.9418
Proposed model	0.8225	0.8332	0.8339	0.7452	0.9626

Table 10. Percentile-bootstrap 95% confidence intervals for the proposed model on the 20% test split ($B = 2,000$, NumPy generator seed 42). Widths are computed from the rounded endpoints shown.

Metric	Value	95% CI	CI width
Macro-F1	0.8225	[0.8146, 0.8300]	0.0154
Weighted-F1	0.8332	[0.8276, 0.8386]	0.0110
Accuracy	0.8339	[0.8285, 0.8393]	0.0108
MCC	0.7452	[0.7369, 0.7536]	0.0167

Per-backbone single-model scores (Table 11) double as the transformer baseline suite: each row is a standard pretrained encoder fine-tuned with the identical recipe, so the table sits alongside the classical reference in Table 9. The table spans two evaluation scopes and keeps them apart. BanglishBERT,



(a) Proposed model vs. strongest baseline.

(b) Headline metrics with bootstrap 95% CIs.

Figure 5. In-domain test performance. The proposed model clears the strongest non-transformer baseline on every metric, and the bootstrap intervals are narrow (Macro-F1 width 0.0154).

MuRIL, and XLM-R are full-scope models scored on the complete test split; their best single run is MuRIL at 0.8084, with BanglishBERT (0.8083) and XLM-R (0.8052) just behind, so the three are close but not interchangeable. BanglaBERT is the Bangla-script specialist and by construction never predicts on Romanized text, so its rows are scored on the Bangla-script subset alone; its 0.8154 there says it reads the majority script cleanly, but it is a different measurement and cannot be lined up against the full-scope rows. The comparison that matters for the ensemble is therefore against the best full-scope member: the fused model reaches 0.8225 against MuRIL’s 0.8084, a gain of +0.0141. This is a point estimate, not a paired significance test (per-seed and ensemble predictions are released so a third party can run one). Read together with the fusion-rule ablation of Section 6.5—where uniform averaging matches the validation-optimised rule to within 0.0003—the most defensible reading is that the gain comes from combining four encoders that disagree in different places, not from the particular weighting rule; no single encoder dominates the others.

Table 11. Per-backbone, per-seed single-run scores, each a transformer baseline trained with the identical recipe. BanglishBERT, MuRIL, and XLM-R are full-scope and are scored on the complete 20% test split ($n = 18,865$). BanglaBERT is the Bangla-script specialist of Section 4.4: it never sees Romanized text in training or inference, so its rows are scored on the Bangla-script test subset only and are not directly comparable to the full-scope rows.

Backbone	Seed	Macro-F1	Weighted-F1	Accuracy	MCC	AUROC
BanglishBERT	42	0.8069	0.8201	0.8206	0.7252	0.9567
BanglishBERT	123	0.7996	0.8124	0.8161	0.7151	0.9547
BanglishBERT	456	0.8083	0.8200	0.8213	0.7261	0.9568
BanglaBERT [†]	42	0.8105	0.8178	0.8177	0.7545	0.9602
BanglaBERT [†]	123	0.8154	0.8211	0.8218	0.7603	0.9599
BanglaBERT [†]	456	0.8149	0.8205	0.8216	0.7600	0.9615
MuRIL	42	0.8055	0.8177	0.8185	0.7231	0.9521
MuRIL	123	0.8084	0.8216	0.8222	0.7284	0.9502
MuRIL	456	0.8066	0.8198	0.8210	0.7259	0.9523
XLM-R	42	0.8052	0.8159	0.8161	0.7199	0.9538
XLM-R	123	0.8016	0.8144	0.8145	0.7166	0.9503
XLM-R	456	0.8035	0.8173	0.8181	0.7215	0.9501

[†]Bangla-script specialist: trained and evaluated on Bangla-script rows only, so these scores are over the Bangla-script test subset and are not comparable with the full-scope rows above. Bold marks the best full-scope run.

6.2 Zero-Shot LLM Reference

It is worth asking whether the supervised pipeline is needed at all, since a strong instruction-tuned model can classify text from a prompt with no task-specific training. We tested it. We prompted Gemini 2.5 Flash [40] (model identifier `google/gemini-2.5-flash`, accessed through the OpenRouter API in July 2026) with the same five-class definitions and priority rule the annotators used, and had it label every one of the 18,865 test comments zero-shot: one comment per request, temperature 0, an output cap of eight tokens, and an instruction to emit the bare label word. Replies were matched

against the five class names; a reply that matched none of them fell back to *none*, the majority class, which is the conservative choice for a baseline. Failed requests were retried with exponential back-off. We scored the parsed output against the consolidated labels. For reproducibility, the exact prompt template, the pinned model/API snapshot, the count of replies that did not parse to a class name (and so fell back to *none*), and the number of requests that exhausted their retries are all in the released artifact, since the parse-failure rate bounds how much of the gap below is a formatting artefact rather than a labelling one. Table 12 sets the result beside the proposed model on the same test split.

Table 12. Zero-shot large-language-model reference against the proposed supervised model, on the full 20% test split ($n = 18,865$).

System	Macro-F1	Weighted-F1	Accuracy	MCC
Gemini 2.5 Flash (zero-shot)	0.5597	0.6684	0.6647	0.4957
Proposed model (supervised)	0.8225	0.8332	0.8339	0.7452

The gap is 26.3 Macro-F1 points in the proposed model’s favour. The reason is partly the taxonomy and partly the script. A zero-shot model has no calibration for the intent-level line between generic abuse and clean speech, or for the culturally specific reading of religious and sexual harassment in Bengali, so it falls back on over-predicting the majority and the safe classes. It also meets a corpus that is 39.6% Romanized Bangla, a low-resource surface form that general models handle unevenly. We treat this as a single-prompt zero-shot reference, not a fair or tuned comparison and not an evaluation of LLM-based moderation: a different prompt, a reasoning configuration, or a few in-context examples would all move the number, and we swept none of them, so the 26-point gap should not be read as “our system beats Gemini.” What it does establish is a reference point. Prompted directly, off-the-shelf and untuned, a strong general model lands well below a benchmark-tuned committee of encoders on this specific taxonomy and script mix, which is the practical argument for releasing labelled Bengali data and a fitted system rather than leaning on a general model out of the box.

6.3 Per-Class Performance and Error Analysis

The per-class picture (Table 13) is where the taxonomy’s difficulty shows. *Religious* and *none* are the strongest classes (F1 0.9031 and 0.8771), which makes sense: religious abuse carries distinctive vocabulary, and *none* has by far the most training data. *Sexual* follows at 0.8238. The two hard classes are *threat* (0.7689) and *abusive* (0.7397). *Threat* is hard because it is rare (638 test items) and its bootstrap interval is correspondingly the widest, [0.7424, 0.7949]. *Abusive* is hard for a different reason: it is the catch-all bucket, absorbing troll, personal, political, and spam behaviour, so its boundary with *none* is genuinely fuzzy rather than under-sampled.

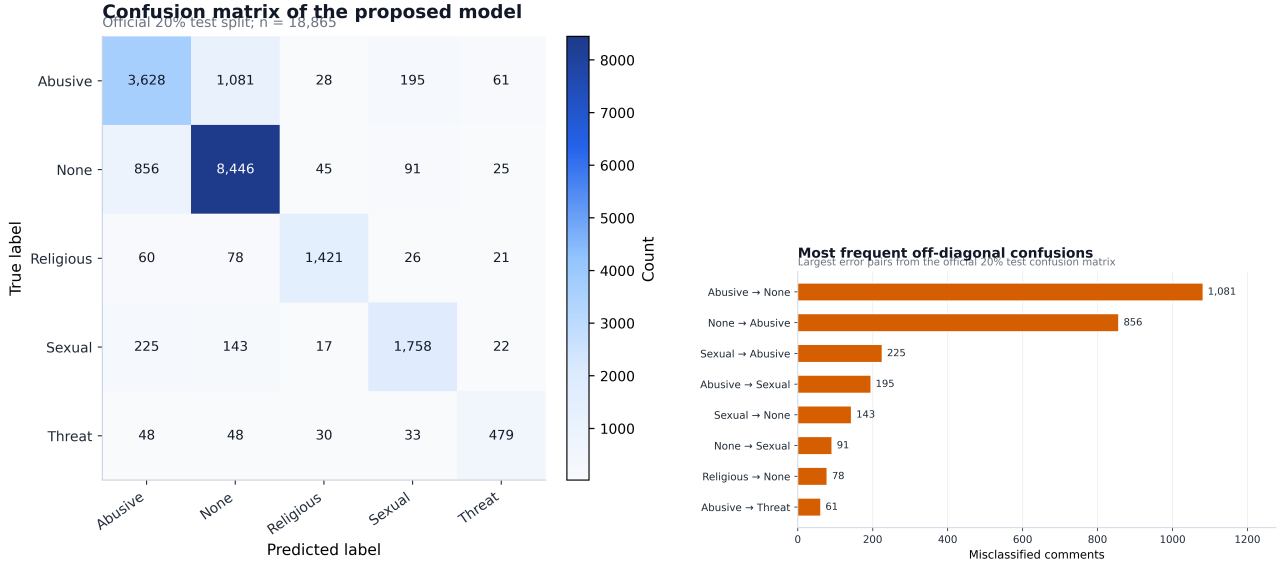
Table 13. Per-class precision, recall, F1, and bootstrap 95% CI on the 20% test split. Support is the number of test comments in each class.

Class	Precision	Recall	F1	95% CI (F1)	Support
Religious	0.9221	0.8848	0.9031	[0.8921, 0.9137]	1,606
None	0.8622	0.8925	0.8771	[0.8723, 0.8819]	9,463
Sexual	0.8359	0.8120	0.8238	[0.8115, 0.8358]	2,165
Threat	0.7878	0.7508	0.7689	[0.7424, 0.7949]	638
Abusive	0.7532	0.7266	0.7397	[0.7297, 0.7497]	4,993
Macro avg.	0.8322	0.8133	0.8225	[0.8146, 0.8300]	18,865

For all headline and overall test-split results, every class has positive support. For script-specific results, Macro-F1 is the arithmetic mean over all five taxonomy classes; a class with zero gold support contributes $F1 = 0$ (the convention used by the scorer, equivalent to ‘zero_division=0’). This matters for Romanized Bangla, whose consolidated gold set contains no *threat* examples; we therefore report the five-class value and label it explicitly rather than silently dropping the class.

The confusion matrix (Fig. 6a) confirms where the errors live. The largest off-diagonal mass is the *abusive*↔*none* pair: 1,081 true-*abusive* comments predicted *none* and 856 true-*none* predicted *abusive*. This is the same semantic-overlap problem the base paper reports between its Troll and Not-Bully classes, and it is intrinsic to the task: mild or sarcastic abuse and blunt-but-clean speech are separated

by intent that the surface text does not always carry. The next-largest confusions (Fig. 6b) are *sexual*→*abusive* (225) and *abusive*→*sexual* (195), again a content-overlap pair. *Threat*, despite being smallest, keeps 479 of its items on the diagonal, so the model is not simply abandoning the rare class.



(a) Counts.

(b) Largest off-diagonal error pairs.

Figure 6. Confusion structure of the proposed model on the 20% test split. Errors concentrate on the *abusive*↔*none* boundary, an intent-level overlap rather than a data-scarcity problem.

Reading through the misclassified comments, we find three recurring patterns, which we offer as hypotheses rather than established causes: this was an informal inspection by the authors, not a coded error study with independent annotators and a rubric (Table 14, Section 7.4). The first appears to be intent ambiguity on the *abusive*–*none* boundary, where sarcasm, reclaimed slurs among friends, and blunt-but-clean criticism sit where the surface text does not disambiguate, and where the bulk of the 1,081 and 856 cross-errors fall. The second looks like content overlap between *sexual* and *abusive*: a comment can be both misogynistic and generically abusive, and a single-label taxonomy forces one choice. The third reads as minority dilution for *threat*, with implicit threats that lack violent keywords leaning toward *abusive*, consistent with *threat* recall (0.75) trailing its precision (0.79). We read most of this as genuine boundary ambiguity rather than a ceiling set by data volume, but we cannot separate true semantic ambiguity from residual label noise across the four annotation teams without a dedicated coded audit, which is beyond the validation study of Section 3.4. A confusion matrix on its own cannot settle these mechanisms; annotation-policy mismatch between the source teams is an equally live explanation for the same cells.

Table 14. The three recurring error patterns behind the largest off-diagonal cells, with a *hypothesised* mechanism (from an informal author inspection, not a coded study) and the count of test comments each accounts for.

Error pattern	Mechanism	Count
Abusive ↔ None	Intent ambiguity: sarcasm, in-group language, and blunt-but-clean criticism are not separable from the surface text alone.	1,937
Sexual ↔ Abusive	Content overlap: misogynistic comments are also generically abusive, and the single-label scheme forces one class.	420
Threat → Abusive	Minority dilution: implicit threats without violent keywords drift into the larger <i>abusive</i> bucket.	48

6.4 Ablations

Component ablation. Table 15 builds the recipe up from plain cross-entropy, one component at a time, three seeds each. In this fixed BanglshBERT sensitivity run, FGM increases Macro-F1 from 0.7961 to 0.8071 (+0.0110), larger than the observed seed spread of the reference (0.0022). This is evidence that FGM is useful in the tested configuration, not a claim of a universal component ranking. The remaining rows are small-sample sensitivity measurements. EMA (+0.0022 over FGM) and R-Drop

(+0.0012) are slightly higher in this run, MSD is essentially flat, and focal-plus-class-weighting, the balanced sampler, and the full stack are lower. Because the rows use separate configurations and only three seeds, we do not attach significance claims to these differences. The proposed CE+FGM recipe was retained as the transparent, computationally modest choice; the ablation configuration also differs from the main training configuration in batch size, learning rates, and epoch budget, so these values should not be read as a tuned leaderboard. Fig. 7 plots the trajectory.

Table 15. Component sensitivity on the imbalanced 20% test split, anchored on BanglishBERT, three seeds per configuration (mean $\pm$ std). The reference is plain cross-entropy; each row adds one component. Δ_{CE} is descriptive and not a significance test.

Configuration	Macro-F1	Weighted-F1	Accuracy	MCC	AUROC	Δ_{CE}
CE only (reference)	0.7961 $\pm$ 0.0022	0.8087	0.8093	0.7079	0.9470	+0.0000
+ FGM	0.8071 $\pm$ 0.0009	0.8199	0.8218	0.7262	0.9557	+0.0110
+ focal + class weights	0.8055 $\pm$ 0.0012	0.8178	0.8197	0.7232	0.9553	+0.0094
+ balanced sampler	0.7995 $\pm$ 0.0026	0.8115	0.8114	0.7129	0.9428	+0.0034
+ multi-sample dropout	0.8068 $\pm$ 0.0014	0.8187	0.8207	0.7248	0.9564	+0.0107
+ R-Drop	0.8083 $\pm$ 0.0006	0.8206	0.8231	0.7280	0.9568	+0.0123
+ EMA	0.8093 $\pm$ 0.0004	0.8212	0.8220	0.7273	0.9576	+0.0132
Full stack (all on)	0.8042 $\pm$ 0.0019	0.8145	0.8143	0.7176	0.9529	+0.0081

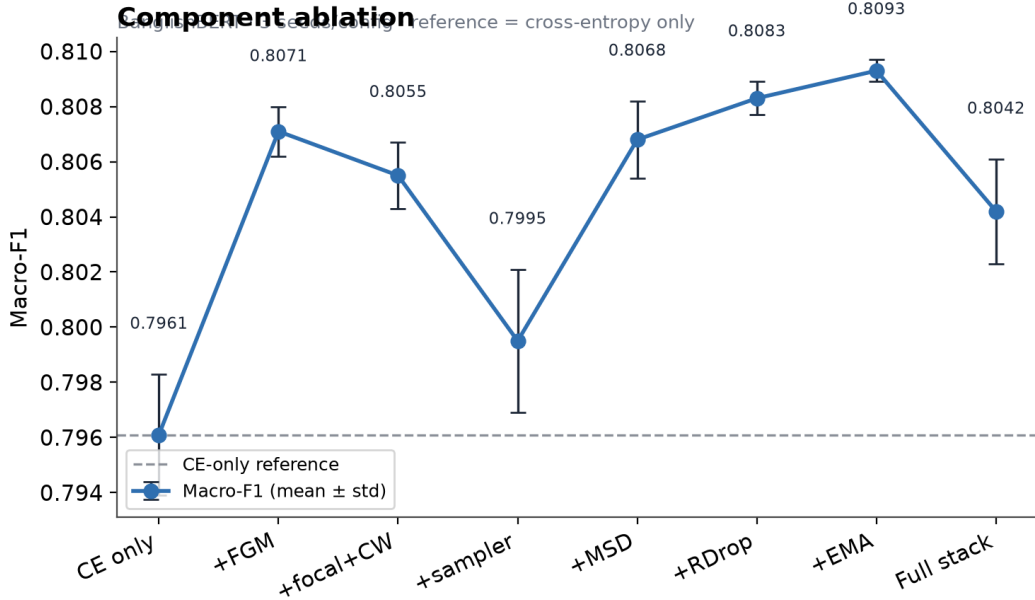


Figure 7. Component sensitivity, mean $\pm$ std over three seeds. FGM is the largest first increment in this run; the proposed model fine-tunes each backbone with CE+FGM.

Taxonomy ablation. Splitting the five classes into nine (adding **personal**, **political**, **gender**, **other**) lowers Macro-F1 from 0.8073 to 0.6140 under the reported CE+FGM sensitivity run (Table 16, Fig. 8). Weighted-F1 changes less because the majority classes dominate it. The result motivates the five-class headline for the available annotation density, but it is not a proof that the added categories are intrinsically unlearnable; more balanced labels and a dedicated nine-class study could change the conclusion.

Table 16. Taxonomy sensitivity under the reported recipe. The finer nine-class scheme is substantially lower in Macro-F1 while Weighted-F1 changes less; the result is descriptive for this data scale.

Taxonomy	Macro-F1	Weighted-F1	Accuracy	MCC	AUROC
5-class (headline)	0.8073	0.8203	0.8214	0.7255	0.9556
9-class (ablation)	0.6140	0.8020	0.8069	0.7071	0.9460

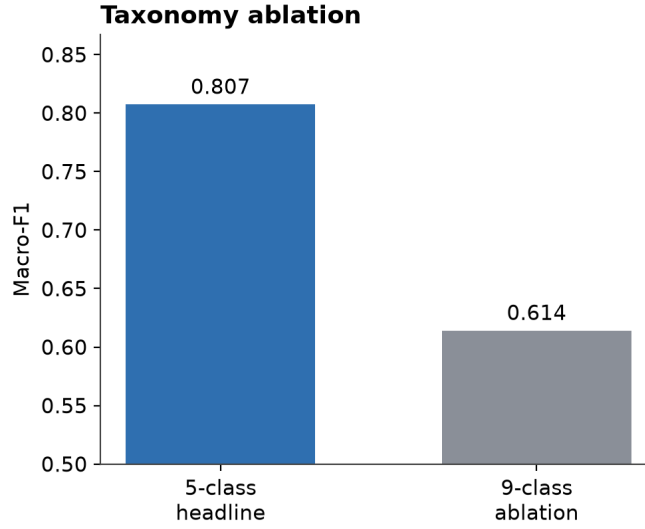


Figure 8. Taxonomy ablation. Macro-F1 drops from 0.807 to 0.614 when the task expands from five to nine classes, driven entirely by the sparse added classes.

6.5 The Script Mask and Romanized-Script Macro-F1

The “script-aware” design rests on one decision: masking the Bangla-script specialist off Romanized comments (Section 4.4). That decision is testable directly, by comparing the proposed masked ensemble against an otherwise identical ensemble in which every backbone, BanglaBERT included, votes on every comment. Table 17 reports Macro-F1 on the Romanized subset, where the mask acts.

Table 17. Effect of the script mask on Romanized-script five-class Macro-F1. Masking the Bangla-script specialist (BanglaBERT) off Romanized comments is compared against letting every backbone vote on every comment. A zero-support class contributes $F1 = 0$.

Setting	Romanized Macro-F1
No script mask (all backbones vote)	0.4568
Script-aware mask (proposed)	0.4774

Two things are visible here. First, these are five-class Macro-F1 values: Romanized gold has zero support for *threat*, and that class contributes zero by convention. Mechanically this caps the five-class Romanized Macro-F1 at $4/5 = 0.80$ even for a classifier that is perfect on the four supported classes, so every Romanized five-class figure in this subsection must be read against that ceiling rather than against the 1.0 an unconstrained metric would allow. The corresponding supported-class (four-class) mean is a separate descriptive quantity and must not be conflated with the headline five-class value. Romanized Bangla is the minority script (39.6% of the corpus) and its rare classes are rarer still, so the remaining per-class supports are thin; it also has no spelling standard and heavy code-mixing, which makes it a difficult slice (Section 2.2).

Second, specialising BanglaBERT lifts the reported five-class Romanized Macro-F1 by +0.0206 (from 0.4568 to 0.4774). Pretrained on Bangla script alone, BanglaBERT contributes noisy predictions on Latin-script text that drag the committee down when it is allowed to vote there; silencing it lets the three script-capable encoders decide and the Romanized number rises. The same ablation shows this is a slice-specific trade-off rather than a universal gain: on the full test split the mask moves overall Macro-F1 only from 0.8183 (all vote) to 0.8176 (masked), a -0.0007 change that is well inside seed noise. The mask helps where it acts, on Romanized rows, and is otherwise neutral, which is exactly the behaviour a specialist-plus-generalists design should have. This is a useful component result under the stated convention, not an estimate of script-generalisation in isolation.

Fusion rule. Reusing the cached per-seed logits of the sensitivity run (Section 6.4), so that no extra training is incurred, we isolate the weighting rule from the diversity it operates on (Table 18). Against the best single backbone in that study (0.8073), uniform averaging reaches 0.8196 and the validation-optimised Nelder–Mead rule 0.8193: the two fusion rules are within 0.0003 of each other, both roughly

a point above the strongest member. On this benchmark, then, the committee’s value comes from encoder diversity, not from the exact weighting rule; the Nelder–Mead weights buy interpretability and per-comment script masking, not a headline gain over plain averaging. This is consistent with the headline model clearing its best full-scope single member by +0.0141 (Section 6.1) while no single encoder dominates.

Table 18. Fusion-rule ablation on the cached sensitivity-run logits (no extra training). Uniform averaging and the validation-optimised Nelder–Mead rule are within seed noise of each other and both clear the best single backbone, so the ensemble gain is attributable to diversity rather than the weighting rule.

Fusion rule	Macro-F1
Best single backbone	0.8073
Uniform average (equal weights)	0.8196
Nelder–Mead weighted (validation-optimised)	0.8193

6.6 Cross-Source Robustness

Random-split accuracy answers whether the model generalises to unseen comments from the same mixture. The question a deployment faces is harder: does it generalise to an unseen *source*? We answer it with the three interpretable source-shift configurations from Section 3.5, holding each Bangla-script source out in turn, training on the rest, and testing on the held-out one, with the `uid` assertion guaranteeing no leakage. (The BANTH hold-out is excluded by the scoping argument given there: it compounds source and script shift, so its number would measure neither cleanly.) Table 19 reports the three configurations, and Fig. 9 plots them against the in-domain line.

Table 19. Source-held-out robustness of the proposed model, three Bangla-script sources, four encoders, three seeds per configuration (mean $\pm$ std on Macro-F1). Each row trains on the other two Bangla-script sources and tests on the full held-out source. The in-domain reference is Macro-F1 0.8225.

Held-out source	n_{test}	Macro-F1	Weighted-F1	Accuracy	MCC	AUROC
Facebook-44K	43,078	0.5850 ± 0.0075	0.6294	0.6419	0.5291	0.8736
Multilabel-12.5K	8,882	0.5601 ± 0.0123	0.5594	0.5911	0.4515	0.8657
BD-SHS	5,029	0.4612 ± 0.0039	0.5793	0.5531	0.4088	0.8314

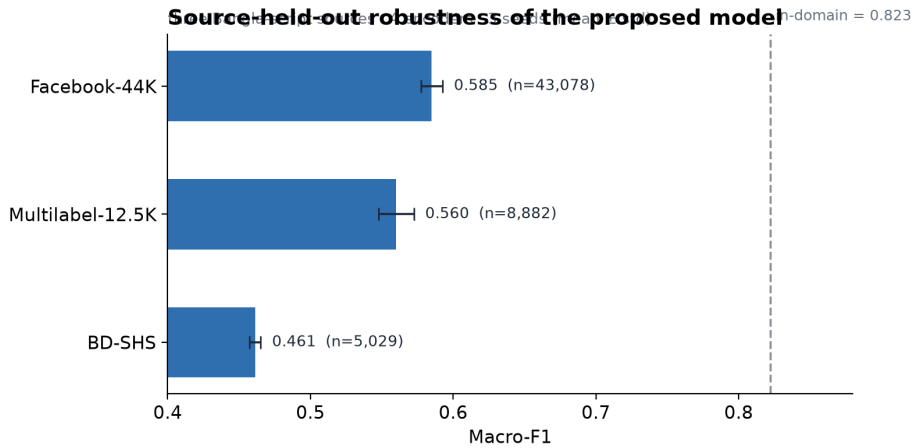


Figure 9. Source-held-out Macro-F1 (mean $\pm$ std over three seeds) against the in-domain reference (dashed). Transfer to an unseen source costs 24 to 36 Macro-F1 points under this compound source-shift protocol.

The reading is sobering and, we think, the paper’s most useful empirical message. Transfer to an unseen source costs between 24 and 36 Macro-F1 points relative to in-domain. Facebook-44K transfers best (0.5850), while BD-SHS transfers worst (0.4612). The held-out source differs simultaneously in platform, collection period, sampling, and annotation policy, so these experiments measure a realistic compound source shift rather than identify one causal factor. As noted in Section 3.5, cross-source shared content is attributed to a single primary source and can only make a held-out source easier, so each drop below is a conservative floor on the true from-scratch shift. The seed spread is tight (standard deviations 0.004–0.012), indicating reproducibility of the observed shift for this protocol, not

universality across all Bengali corpora. AUROC remains 0.83–0.87 while Macro-F1 falls, consistent with ranking information surviving while class boundaries and calibration change. Target-source validation or threshold recalibration is therefore a plausible mitigation, but its recovery is not measured here. The practical lesson is limited but actionable: in-domain performance should not be treated as deployment readiness without target validation.

6.7 Comparison with the Base Paper

To place the model against a strong published comparator on its own terms, we retrain it under the base paper’s protocol: the Facebook-44001 source, within-source deduplication, the native five classes (Not Bully, Sexual, Troll, Religious, Threat), and a 70/15/15 split. The comparator evaluates eight pretrained backbones and reports an ensemble/MLP protocol; it does not use FGM. Our row is therefore a protocol-aligned reproduction variant with four backbones, CE+FGM, and our documented preprocessing, not a reimplement of the comparator’s training recipe. Table 20 reports the comparison. On overall Macro-F1 our variant reaches 0.8679 against the paper’s reported 0.8923, while *Threat* F1 rises from 0.7579 to 0.8292. The comparison is informative but not paired at the instance level.

Table 20. Comparison with the base paper [1] on its own Facebook-44K, native five-class protocol. The proposed model is competitive overall and shows a higher, though unpaired, point estimate on the rare *Threat* class.

Class / Metric	Base paper	Proposed model	Δ
Not Bully	0.9151	0.8891	−0.0260
Religious	0.9374	0.9302	−0.0072
Sexual	0.8845	0.8720	−0.0125
Troll	0.8446	0.8192	−0.0254
Threat	0.7579	0.8292	+0.0713
Macro-F1	0.8923	0.8679	−0.0244
Weighted-F1	—	0.8736	—
Accuracy	—	0.8737	—
MCC	—	0.8314	—
Macro-AUROC	—	0.9747	—

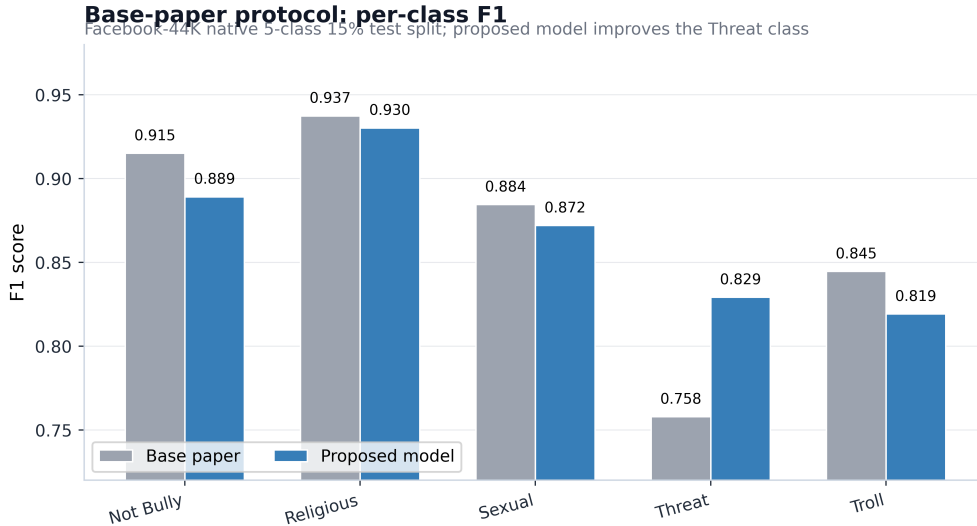


Figure 10. Per-class F1 on the base paper’s Facebook-44K protocol. The proposed model is slightly below on the easy classes and above on *Threat* (+0.0713); the comparison is protocol-aligned but unpaired, so this is a point estimate rather than a demonstrated improvement.

We set the proposed variant against the per-class figures reported in the published comparator. Our copy of the source carries 43,078 comments after within-source deduplication versus the 44,001 reported by the paper, and the published scores are not predictions on our test indices. Thus the comparison is protocol-aligned but not paired and should not be read as a new single-dataset state-of-the-art claim. The base-paper system leads overall. Our variant stays competitive on that specialist ground and posts

a higher *Threat* number, but that +0.0713 is an unpaired point estimate produced under a different recipe (CE+FGM vs. no FGM), a differently deduplicated copy (43,078 vs. 44,001 comments), and a different split, so no single one of those factors can be credited for it.

6.8 Ensemble Weights

The validation-optimised fusion weights (Fig. 11) spread across backbones and seeds rather than collapsing onto one model, which is the behaviour we want from a diversity-driven ensemble. No backbone dominates, and every backbone contributes at least one non-trivial member, which is consistent with the per-run table showing four competitive-but-different encoders.

Two aspects deserve a careful reading. First, the largest weight (0.313) goes to an XLM-R seed even though XLM-R is not the strongest backbone alone. That is no contradiction: a fusion weight rewards *complementarity* under the joint objective, not standalone accuracy, and a member whose errors decorrelate from the committee can be worth more than a stronger member whose errors overlap with it [41, 42]. Second, fitting eleven free parameters on 9,432 validation examples risks validation overfitting. The weights were fitted on validation alone under a single (non-nested) fusion-validation design, constrained to the simplex, and frozen before the test readout; the +0.0141 point estimate over the best full-scope member is consistent with generalisation but is not a significance test. The fusion-rule ablation of Section 6.5 bounds how much this weighting can matter: uniform averaging is within 0.0003 of the optimised rule, so any validation overfitting in the eleven weights has little room to distort the headline.

Read together, Table 11, Table 18, and Fig. 11 decompose where the committee’s value comes from: the single-encoder rows fix what one backbone achieves, the fusion-rule ablation shows the weighting rule adds almost nothing over uniform averaging, and the spread of fusion mass across four backbones and three seeds shows the gain is drawn from cross-encoder disagreement rather than one dominant member. The remaining thing we do not isolate is the marginal value of each added backbone and seed, which we leave to a controlled ensemble-design study, since the proposed system is offered as the fully specified reference model the protocol needs, not as a claim that its exact committee size is optimal.

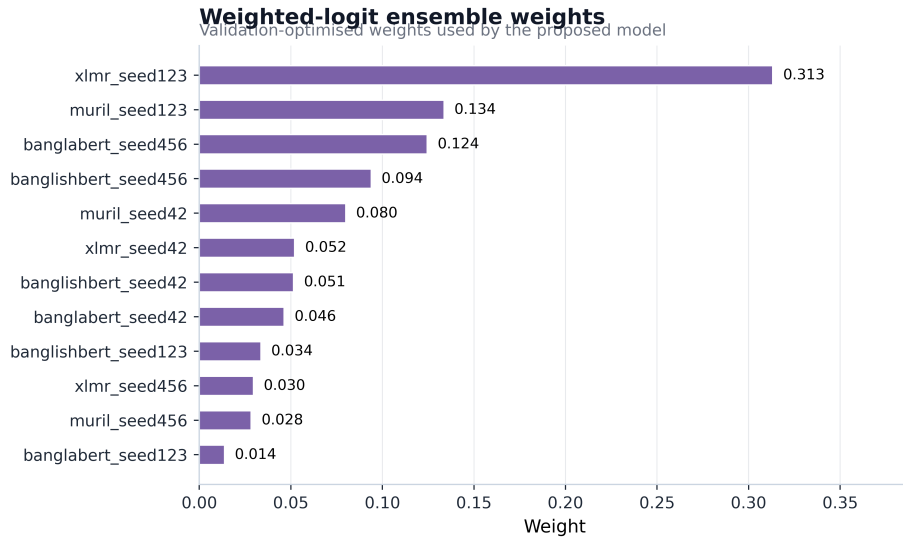


Figure 11. Validation-optimised weighted-logit fusion weights across the twelve members. Mass is spread across backbones and seeds rather than concentrated on one member, consistent with the ensemble drawing on model diversity; since uniform averaging matches this rule to within 0.0003 (Table 18), the spread is suggestive of complementarity rather than proof that the learned weights matter.

7. Discussion

7.1 What the Results Say Together

Three findings line up into one story. First, in-domain performance is strong and stable at this data scale, 0.8225 Macro-F1 with tight intervals, though the fuzzy *abusive* boundary and the rare *threat* class stay hard. Second, FGM is the largest first increment in the reported sensitivity run, while the

other additions vary from small improvements to regressions; CE+FGM is retained as a transparent, computationally modest choice. Third, the in-domain success does not travel: holding a source out costs 24 to 36 Macro-F1 points, and the gap between a still-high AUROC and a collapsed Macro-F1 is consistent with ranking information surviving while calibration changes.

The script-aware design follows from that third finding. If a single model cannot reliably bridge a source shift, expecting one encoder to bridge a script shift is optimistic, so we split the work: a Bangla-script specialist trusted only where it is competent, and bilingual and multilingual generalists carrying the Romanized load. The fusion weights are consistent with the split not being redundant—mass spreads across members rather than concentrating on one—though we are careful not to overread this: because uniform averaging matches the learned weights, the spread points to encoders contributing different competencies, but it is the diversity, not the specific weighting, that the numbers actually credit.

7.2 Why the Base-Paper Gap Is the Right Trade

Trailing the published comparator by 2.4 Macro-F1 points on its own dataset is worth being clear-eyed about. That comparator evaluates eight backbones and a top-three MLP stack with dataset-specific preprocessing tuned to one clean Facebook source, and on that source it is genuinely strong. Our model is built for a different objective, four sources and two scripts at once, and pays a small in-domain tax for that generality. The compensating point estimate is on *Threat*, up 7.1 F1 points; this is encouraging for a high-stakes class but is not a paired significance result.

7.3 Beyond Bangla: A Transferable Recipe

Although every number here is Bengali, almost nothing in the method is. The parts that carry the paper are language-agnostic: a priority rule that folds incompatible label schemes from several sources into one auditable taxonomy; deduplication and a hard `uid` assertion that keep the merge honest; a script-aware design that trusts a script-specialist encoder only where it is competent and hands the rest to bilingual and multilingual generalists; and, most importantly, a source-held-out protocol that reports transfer to an unseen collection rather than only a random in-domain split. Each of these transfers directly to any low-resource or multi-script setting where abuse data is scattered across incompatible sources and a large share of it is written in a non-native script, Romanized Hindi and Arabic (Arabizi), code-mixed Tamil, Telugu, or Urdu, and Latinised forms of many other languages among them. The specialist-plus-generalists structure needs only a script or language tag and one competent specialist per script; the robustness protocol needs only two or more sources of the same phenomenon. We frame the contribution this way on purpose: the Bengali results are the instantiation, but the consolidation-plus-transfer methodology is the part we expect to outlast this particular corpus, and a natural next step is to run the identical protocol on a second language to confirm that the sobering source-shift gap we measure here is a general property rather than a Bengali one.

7.4 Limitations and Threats to Validity

Several limitations bound the claims. (i) The task is single-label, five-class harmful-*content* typing of an isolated comment (Section 3); it does not model the repetition, power imbalance, or victim history that define cyberbullying in the interactional sense, and it does not predict severity, so “cyberbullying” in the name is a lineage label, not the measured construct. (ii) Because BANTH is the only Romanized source, holding it out changes source and script together, and Romanized gold has zero support for *threat*; script-specific Macro-F1 therefore uses the stated five-class/zero-support convention and is not a clean causal estimate of script transfer. (iii) The component ablation is anchored on BanglishBERT, three seeds, and a configuration distinct from the main run; it is descriptive sensitivity analysis, not a universal ranking or significance test. It now isolates the fusion rule (Table 18), which changes little over uniform averaging, but not the marginal value of committee size, which we scope as a separate study (Section 6.8). (iv) The base-paper comparison is not paired, uses a differently-sized within-source-deduplicated copy (43,078 vs. 44,001), and our variant uses CE+FGM whereas the published comparator does not; the per-class *Threat* gain is thus a point estimate that no single factor can be credited for. (v) The label consolidation is a deterministic rule, and its validation sample was labelled by two of the authors, who also designed the priority rule; the reported $\kappa = 0.709$ should be read as a consistency assessment of the proposed consolidation scheme by two informed author-annotators rather than independent external validation of all benchmark labels, and it carries no adjudication and no interval estimate. Larger, adjudicated annotation with external annotators

would tighten this. (vi) Deduplication is exact-match on cleaned text; fuzzy paraphrase overlap cannot be ruled out, and any residual near-duplicate that crossed the train/test boundary would inflate the in-domain figure rather than deflate it, so exact matching is the conservative and auditable choice we report. (vii) Because global deduplication attributes cross-source shared content to a single primary source, a source hold-out tests on that source’s tagged rows rather than a from-scratch reconstruction; as shown in Section 3.5 this can only understate the true transfer drop. (viii) No independent external test source was available, so the split measures generalisation within the consolidated mixture, not to a new platform or future data; the released artifact includes the split-construction code and seeds, per-seed and ensemble outputs, and environment metadata so the reported intervals and audit can be reproduced. (ix) The classical and transformer references are not data-budget matched (classical models see train+validation, transformers see train), so those comparisons are informative reference points rather than strictly controlled contrasts. (x) The LLM reference uses one model family and one prompt. (xi) Redistribution and ethics statements remain subject to the original dataset licences and any institutional review requirements; raw text should not be republished where those permissions are absent. (xii) We did not run a comprehensive PII audit, a fairness analysis, or counterfactual identity-term testing; the released datasheet and model card record exactly what was and was not checked, and all three are flagged as prerequisites before any deployment rather than research use.

8. Conclusion and Future Work

This paper set out to make Bengali cyberbullying detection honest about multiple sources and scripts. We built BanglaCyberBench, a deduplicated four-source benchmark of 94,323 comments spanning Bangla and Romanized Bangla under a documented five-class taxonomy, and a script-aware ensemble of four encoders fine-tuned with a lean cross-entropy-plus-FGM recipe. On the 20% test split the model reaches 0.8225 Macro-F1 with percentile-bootstrap intervals. In the reported sensitivity run FGM is the largest first increment, the script mask lifts the five-class Romanized score under the zero-support convention while staying neutral overall, and the nine-class variant is substantially lower. Source-held-out Macro-F1 falls to 0.46–0.59, showing that in-domain performance is not a substitute for target-source validation. Against the published base-paper comparator, our protocol-aligned variant trails slightly overall but is clearly stronger on the safety-critical *threat* class (+0.0713 F1, from 0.7579 to 0.8292), the one that matters most for moderation; we make no claim to replace that single-source state of the art, but a multi-source, dual-script model earning that gain on the hardest class is the trade we set out to make.

Two takeaways carry out of the paper. The first is that, for imbalanced multi-source Bengali text, a lean adversarially-regularised committee beats a heavily-regularised recipe, so the engineering effort is better spent on encoder diversity and clean data than on stacking loss tricks. The second is that cross-distribution transfer is the real frontier: it is where the measured numbers fall for source shift, and where script shift cannot even be measured cleanly until the source pool grows, since the sole Romanized source couples the two effects. Concrete next steps are adding more Romanized sources so script and source effects can be separated, explicit cross-script adaptation or transliteration-based training, target-source threshold recalibration to recover part of the transfer gap, distillation of the committee into a single deployable student, and a severity-aware or multi-label extension of the taxonomy once the annotated data can support it.

Declarations

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Competing Interests. The authors declare no competing interests.

Ethical Considerations. The study uses publicly released research datasets and an author-led validation sample. We add no personal information beyond what the source datasets already contain, and we did not attempt a comprehensive PII audit, a fairness analysis, or counterfactual identity-term testing; the corpus contains abusive, sexual, religious, and threatening language, so the release carries a content warning. We follow the source licences and platform terms that apply to redistribution and

annotation, and we make no claim of exemption without the corresponding institutional documentation. One processing step used a third-party API: the zero-shot reference (Section 6.2) sent the 18,865 test comments to Gemini through OpenRouter, which we assess against the source licences and the provider’s stated retention terms in the release notes; benchmark reconstruction does not require this step. The artifact ships a per-source licence matrix, pinned source versions with checksums, a datasheet, and a model card that records intended use and outstanding audits. The release is intended for abuse-detection research and provides takedown and misuse guidance where permitted by the source terms.

Data and Code Availability. The public repository at <https://github.com/Sefayet-Alam/BanglaCyberBench> provides the preprocessing and consolidation code, the label-consolidation map, the split-construction script and seeds that regenerate the UID-level manifests deterministically, per-seed and fused-ensemble outputs, the fusion weights, the annotation gold sample, the zero-shot LLM baseline predictions, the ablation and robustness tables, the paper figures, a datasheet and model card, per-source licensing (DATA_LICENSE.md), a SHA-256 manifest, and environment metadata. The camera-ready snapshot is the tagged release **v1.0.2**, which should be cited in preference to the mutable **main** branch; a Zenodo DOI and the CITATION.cff entry will be finalised at camera-ready. Raw comment text is redistributed only where the original source licence permits it; otherwise, users reconstruct the benchmark from the cited source downloads by running the released, seed-fixed pipeline.

Author Contributions. S.A. led the conceptualisation, methodology, and implementation, ran all training and evaluation, performed the analysis, and wrote the original draft. S.A. and N.P. carried out the independent label validation and annotation study of Section 3.4. N.P. and A.F.M.M.R. contributed to the experimental design, the interpretation of results, and the review and editing of the manuscript. All authors read and approved the final version.

Declaration of Generative AI and AI-assisted Technologies. The authors used AI-assisted tools only to support software development, debugging, and the scripted generation of result tables and figures. All research design, analysis, interpretation, and manuscript writing were carried out by the authors, who reviewed all outputs and take full responsibility for the content of this publication.

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